

Project Data Page:



Table of Contents

Cover Page	1
Title Page	2
Vessel Design Summary:	3
Nozzle Summary:	6
Nozzle Schedule:	7
Bill of Material:	9
MDMT Summary:	10
Warnings and Errors:	12
Input Echo:	13
XY Coordinate Calculations:	23
Internal Pressure Calculations:	24
Element: EH-1	25
Element: Shell-1	27
Element: Shell-2	29
Element: Shell-3	31
Element: Shell-3	33
Element: Shell-4	35
Element: EH-2	37
Shop Pressure Test Results:	39
Field Pressure Test Results:	41
External Pressure Calculations:	43
Element and Detail Weights:	44
Nozzle Flange MAWP:	46
Wind Load Calculation:	48
Earthquake Load Calculation:	50
Center of Gravity Calculation:	52
Saddle Calcs: Operating Case:	53
Saddle Calcs: Test Case:	68
Nozzle Calcs.: M1-24''	82
Nozzle Calcs.: K2A-2''	88
Nozzle Calcs.: N3-6''	91
Nozzle Calcs.: N4-4''	105
Nozzle Calcs.: K2B-2''	119
Nozzle Calcs.: K3-2''	122
Nozzle Calcs.: K4-2''	125
Nozzle Calcs.: N5-4''	128
Nozzle Calcs.: N1-28''	142
Nozzle Calcs.: N2-4''	156
Nozzle Calcs.: N6-4''	160
Nozzle Calcs.: K1A-2''	174
Nozzle Calcs.: K1B-2''	177
Nozzle Calcs.: M2-24''	180



Expansion vessel (Electrolyzer Area)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A162-V-XX01-Expansion vessel (Electrolyzer Area)-7%.pvdb

Date of Analysis : Sep 18, 2026 9:39am

PV Elite 28, January 2026



Customer : by:
item : date:
Item No : S/O:

Design Internal Pressure : Temp. :
Design External Pressure : Temp. :

Head Matl.:	Corr. All.:	JE :
Shell Matl.:	Corr. All.:	JE :
Flange Matl.:	Corr. All.:	JE :
Pipe Matl.:	Corr. All.:	JE :
Cone Matl.:	Corr. All.:	JE :

Radiographic Requirements :

Post Weld Heat Treat:

Wind Specification : V : Exp.: Imp :

Seismic : Zone : Cat.:

Results ...

Basic Flange Class: Rating : Ambient:

MAWP (Corroded) : limited by :
MAP (New & Cold): limited by :
Min. Design Metal Tmp: without Impacts
Charpy Impact Reqmnts.:

Stiffening Rings Required :

hydrotest Requirement :

Notes ...

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 4400.000 mm ID	
Vessel Design Length, Tangent to Tangent	13300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	1.000 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	1.08 MPa
Shop Test Pressure	1.400 MPa
Field Test Pressure	1.400 MPa
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	-14.0 °C
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
EH-1	1.035	0.00	1.080	3.0000	No	No
Shell-1	1.035	0.00	1.080	3.0000	N/A	No
Shell-2	1.035	0.00	1.080	3.0000	N/A	No
Shell-3	1.035	0.00	1.080	3.0000	N/A	No
Shell-3	1.035	0.00	1.080	3.0000	N/A	No
Shell-4	1.035	0.00	1.080	3.0000	N/A	No
EH-2	1.035	0.00	1.080	3.0000	No	No

Liquid Level: 3520.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	28.0	24.0	22.4	...	0.85	0.85
Cylinder	3050.0	3000.0	24.0	24.0	22.5	...	0.85	0.85
Cylinder	6050.0	3000.0	24.0	24.0	22.5	...	0.85	0.85

Cylinder	7250.0	1200.0	24.0	24.0	22.5	...	0.85	0.85
Cylinder	10250.0	3000.0	24.0	24.0	22.5	...	0.85	0.85
Cylinder	13250.0	3000.0	24.0	24.0	22.5	...	0.85	0.85
Ellipse	13300.0	50.0	28.0	24.0	22.4	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-6''	WRC-297	0.819	Shell	Passed
N4-4''	WRC-297	0.819	Shell	Passed
N5-4''	WRC-297	0.819	Shell	Passed
N1-28''	WRC-297	0.848	Shell	Passed
N6-4''	WRC-297	0.819	Shell	Passed

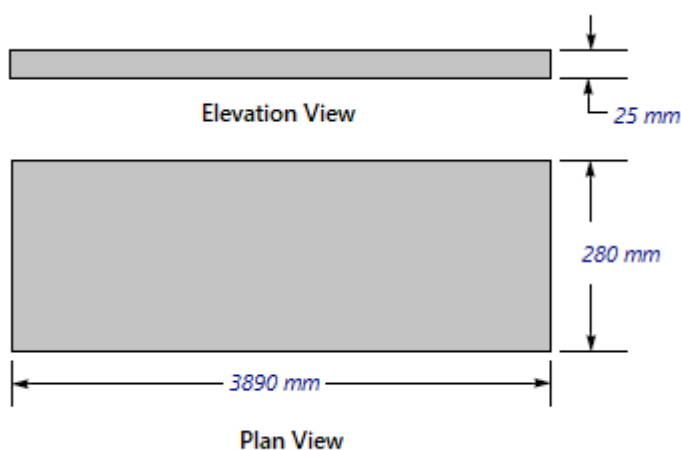
Foundation/Support Summary

Saddle Parameters:

Saddle Width	258.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	2500.000 mm
Wear Pad Width	330.000 mm
Wear Pad Thickness	22.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1200.000 mm

Baseplate Length	3890.000 mm
Baseplate Thickness	25.000 mm
Baseplate Width	280.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	22.000 mm
Web Thickness	22.000 mm
Height of Center Web	251.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	1523509.62 N
Maximum Transverse Saddle Shear Load	119748.73 N
Maximum Longitudinal Saddle Shear Load	239497.45 N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1458320.00 N
Maximum Transverse Saddle Shear Load	19994.16 N
Maximum Longitudinal Saddle Shear Load	16222.51 N

**Weights:**

Fabricated - Bare w/o removable internals	53854.9	kgm
Shop Test - Fabricated + water (full)	278289.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	53854.9	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	64888.7	kgm
Empty - Fab. + intls. + details + wghts.	64888.7	kgm
Operating - Empty + operating liquid (no ca)	258282.8	kgm
Field Test - Empty weight + test medium (full)	289323.5	kgm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Nozzle Calculation Summary (MPa & mm):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1-24''	1.08	...	...	...		OK	Passed
K2A-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
N3-6''	1.08	...	...	OK	9.22	OK	Passed
N4-4''	1.08	...	...	OK	8.26	OK	Passed
K2B-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
K3-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
K4-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
N5-4''	1.08	...	...	OK	8.26	OK	Passed
N1-28''	1.08	...	...	OK	11.33	OK	Passed
N2-4''	5.75	...	...	OK	6.21	OK	No Calc[*]
N6-4''	1.08	...	...	OK	8.26	OK	Passed
K1A-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
K1B-2''	1.08	...	...	OK	6.42	OK	No Calc[*]
M2-24''	1.08	...	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1.08 Nozzle : K1B-2''
 Minimum MAWP Shells/Flanges : 1.08 Element : Shell-4
 Minimum MAPnc Shells/Flanges : 1.27 Element : Shell-4

Computed Vessel M.A.W.P. : 1.08 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1-24''	0.000	0.000	1183.200
20	K2A-2''	550.000	90.000	92.800
20	N3-6''	1550.000	90.000	310.145
20	N4-4''	2950.000	90.000	201.662
20	K2B-2''	550.000	270.000	92.800
30	K3-2''	4050.000	90.000	92.800
30	K4-2''	5550.000	90.000	92.800
40	N5-4''	6650.000	90.000	201.662
40	N1-28''	6650.000	270.000	1354.400
40	N2-4''	0.000	180.000	167.200
60	N6-4''	11450.000	90.000	196.099
60	K1A-2''	12450.000	90.000	92.800
60	K1B-2''	12450.000	270.000	92.800
70	M2-24''	0.000	0.000	1183.200

The nozzle spacing is computed by the following:

$$= \sqrt{(l^2 + l_c^2)} \text{ where}$$

l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2A-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	324.15	150
K2B-2''	2.000 in	Actual	WNF	2.000	10.000	...	...	324.15	150
K3-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	324.15	150
K4-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	324.15	150
K1A-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	324.15	150
K1B-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	324.15	150
N4-4''	4.000 in	120	WNF	4.500	11.125	200.00	24.00	324.74	300
N5-4''	4.000 in	120	WNF	4.500	11.125	200.00	24.00	324.74	300
N2-4''	4.000 in	Actual	WNF	4.000	12.000	200.00	22.00	223.87	150
N6-4''	4.000 in	120	WNF	4.500	11.125	200.00	24.00	324.74	300
N3-6''	6.000 in	80	WNF	6.625	10.973	300.00	24.00	325.61	300
M1-24''	24.000 in	Actual	WNF	24.000	12.000	1000.00	24.00	335.69	150
M2-24''	24.000 in	Actual	WNF	24.000	12.000	1000.00	24.00	335.69	150
N1-28''	28.000 in	Actual	WNF	28.000	20.000	1300.00	24.00	452.93	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2A-2''	SA-106 B	22.000	10.000	...	...	...
K2B-2''	SA-106 B	22.000	10.000	...	...	...
K3-2''	SA-106 B	22.000	10.000	...	...	...
K4-2''	SA-106 B	22.000	10.000	...	...	...
K1A-2''	SA-106 B	22.000	10.000	...	...	...
K1B-2''	SA-106 B	22.000	10.000	...	...	...
N4-4''	SA-106 B	22.000	10.000	18.000	22.000	...
N5-4''	SA-106 B	22.000	10.000	18.000	24.000	...
N2-4''	SA-106 B	22.000	10.000	18.000	22.000	...
N6-4''	SA-106 B	22.000	10.000	18.000	22.000	...
N3-6''	SA-106 B	22.000	10.000	18.000	22.000	...
M1-24''	SA-516 70	22.000	10.000	18.000	24.000	...
M2-24''	SA-106 B	22.000	10.000	18.000	22.000	...
N1-28''	SA-106 B	22.000	10.000	18.000	24.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2A-2''	550.000	90.0	300.00	0.00	Shell-1
K2B-2''	550.000	270.0	300.00	0.00	Shell-1
K3-2''	4050.000	90.0	300.00	0.00	Shell-2
K4-2''	5550.000	90.0	300.00	0.00	Shell-2
K1A-2''	12450.000	90.0	300.00	0.00	Shell-4
K1B-2''	12450.000	270.0	300.00	0.00	Shell-4
N4-4''	2950.000	90.0	300.00	0.00	Shell-1
N5-4''	6650.000	90.0	300.00	0.00	Shell-3

Expansion vessel (Electrolyzer Area)

PV Elite 28 Licensee: SL User



FileName : A162-V-XX01-Expansion vessel (Electrolyzer Area)- Page 8 of 185

Nozzle Schedule: Step: 36 9:39am Sep 18,2026



HEXAGON

N2-4''	6300.000	180.0	200.00	0.00	N1-28''
N6-4''	11449.999	90.0	300.00	0.00	Shell-4
N3-6''	1550.000	90.0	300.00	0.00	Shell-1
M1-24''	...	0.0	300.00	0.00	EH-1
M2-24''	...	0.0	300.00	0.00	EH-2
N1-28''	6650.000	270.0	400.00	0.00	Shell-3

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm THK X 4400.0mm ID X 50.0mm	SA-516 70
4	CYLINDER: 24.0mm THK X 4400.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 24.0mm THK X 4400.0mm ID X 1200.0mm	SA-516 70
2	SADDLE: 258.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1000mm X 4448mm WIDE	
1	LADDER: 5000.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
EH-1	[10]	B	-2	-11	-29	0.844	24.000	0.85	No
EH-1	[7]	B	2	-14		0.714	28.000	0.85	No
Shell-1	[8]	B	-2	-10	-29	0.850	24.000	0.85	No
Shell-2	[8]	B	-2	-10	-29	0.850	24.000	0.85	No
Shell-3	[8]	B	-2	-10	-29	0.850	24.000	0.85	No
Shell-3	[8]	B	-2	-10	-29	0.850	24.000	0.85	No
Shell-4	[8]	B	-2	-10	-29	0.850	24.000	0.85	No
EH-2	[10]	B	-2	-11	-29	0.844	24.000	0.85	No
EH-2	[7]	B	2	-14		0.714	28.000	0.85	No
Left Saddle	[24]	B	-5	-13	-29	0.850	22.000	0.85	No
Left Saddle	[24]	B	-29	-37	-29	0.850	10.000	0.85	No
M1-24''	[1]	B	-2	-29	-29	0.745	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
K2A-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
N3-6''	[1]	B	-2	-29	-29	0.823	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-4''	[1]	B	-2	-29	-29	0.823	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B-2''	[1]	B	-8	-104	-29	0.034	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-43					
K3-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
K4-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
N5-4''	[1]	B	-2	-29	-29	0.823	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-28''	[1]	B	-2	-29	-29	0.849	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-43					
N2-4''	[1]	B	-3	-104	-29	0.053	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-43					
N6-4''	[1]	B	-2	-29	-29	0.823	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
K1B-2''	[1]	B	-29	-104	-29	0.034	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-43					
M2-24''	[1]	B	-2	-29	-29	0.745	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: 2 -10

Required Minimum Design Metal Temperature -10.0 °C

Warmest Computed Minimum Design Metal Temperature -14.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Units used in this Analysis (SIASME):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Expansion vessel (Electrolyzer Area)

Design Internal Pressure (for Hydrotest)	1	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-4	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm
Distance Upwind of Crest	Lh 0 mm
Distance from Crest to the Vessel	x 0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes



Consider Longitudinal Stress in MDMT Calculations

Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	EH-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	4400 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-24''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	4490.3 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	10
Detail Type	Weight
Detail ID	MH DAVIT-1
Dist. from "FROM" Node / Offset dist	-1000 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder



Description	Shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4400 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1150 mm
Width of Saddle	258 mm
Height of Saddle at Bottom	2500 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	330 mm
Thickness of Wear Plate	22 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	20
Detail Type	Platform
Detail ID	PF
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	19931600 mm ²
Platform Weight	101119 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	4448 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.3
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	5000 mm
Unit Weight of Ladder	0.49 N/mm
Platform Length (top head platform)	13300 mm
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm ³
Element From Node	20
Detail Type	Nozzle
Detail ID	K2A-2''
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150



Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.427	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N3-6''	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	553	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N4-4''	
Dist. from "FROM" Node / Offset dist	2900	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	286.77	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2B-2''	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.427	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	1274.8	N
Offset from Element Centerline	0	mm
Element From Node	20	
Detail Type	Weight	
Detail ID	LL-1	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	7207.4	N
Offset from Element Centerline	2200	mm

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4400	mm
Element Thickness	24	mm



Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3520	mm
Liquid Density	0.0009984	kg/cm³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-2''	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.427	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K4-2''	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.427	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	1200	mm
Inside Diameter	4400	mm
Element Thickness	24	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	



Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	N5-4''
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.77 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	N1-28''
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	28 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	3861.1 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	N2-4''
Dist. from "FROM" Node / Offset dist	250 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	174.56 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4400 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No



Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm ³

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4400 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1850 mm
Width of Saddle	258 mm
Height of Saddle at Bottom	2500 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	330 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	N6-4''
Dist. from "FROM" Node / Offset dist	1200 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	253.2 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
-------------------	----



Detail Type	Nozzle
Detail ID	K1A-2''
Dist. from "FROM" Node / Offset dist	2200 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	61.427 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Nozzle
Detail ID	K1B-2''
Dist. from "FROM" Node / Offset dist	2200 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	61.232 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	LL-2
Dist. from "FROM" Node / Offset dist	2000 mm
Miscellaneous Weight	7207.4 N
Offset from Element Centerline	2200 mm

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	EH-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	4400 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3520 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	70
Detail Type	Nozzle
Detail ID	M2-24''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None

Expansion vessel (Electrolyzer Area)

PV Elite 28 Licensee: SL User



HEXAGON

FileName : A162-V-XX01-Expansion vessel (Electrolyzer Area)- Page 22 of 185

Input Echo: Step: 1 9:39am Sep 18,2026

Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4490.3	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	70	
Detail Type	Weight	
Detail ID	MH Davit-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	EH-1	50	...	50	...
	Shell-1	3050	...	3000	...
	Shell-2	6050	...	3000	...
	Shell-3	7250	...	1200	...
	Shell-3	10250	...	3000	...
	Shell-4	13250	...	3000	...
	EH-2	13300	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	EH-1	1.0345	28	3	4400	117.3	1
	Shell-1	1.0345	24	3	4400	117.3	2
	Shell-2	1.0345	24	3	4400	117.3	3
	Shell-3	1.0345	24	3	4400	117.3	4
	Shell-3	1.0345	24	3	4400	117.3	5
	Shell-4	1.0345	24	3	4400	117.3	6
	EH-2	1.0345	28	3	4400	117.3	7

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	EH-1	1	1.08	1.28	24	22.4118	1
	Shell-1	1	1.08	1.27	24	22.5332	2
	Shell-2	1	1.08	1.27	24	22.5332	3
	Shell-3	1	1.08	1.27	24	22.5332	4
	Shell-3	1	1.08	1.27	24	22.5332	5
	Shell-4	1	1.08	1.27	24	22.5332	6
	EH-2	1	1.08	1.28	24	22.4118	7

Minimum

1.08

1.27

MAWP: 1.08 MPa, limited by: Shell-4.

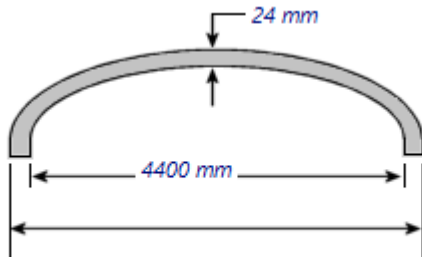
Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-1

Elliptical Head Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress		127.6 MPa
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		22223490.0 mm²	SG of Contents		1.000
Element Volume		11913.0 ltr	Weight of Contents		104605.4 N
Empty Weight		51418.7 N	Operating Weight		156024.1 N
			Comp. All. Ope. Stress		108.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 55.000 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.03 \cdot 4406 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 1.03)$$

$$= 19.4118 + 3.0000 = 22.4118 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 21) / (1 \cdot 4406 + 0.2 \cdot 21)$$

$$= 1.12 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 24) / (1 \cdot 4400 + 0.2 \cdot 24)$$

$$= 1.28 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.03 \cdot (1 \cdot 4406 + 0.2 \cdot 21)) / (2 \cdot 0.85 \cdot 21)$$

$$= 127.570 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03) + 3$
 $= 22.533 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.035 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 25) / (2203 + 0.6 \cdot 25)$
 $= 1.32 - 0.035 = 1.29 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (4406 / (2 \cdot 1103)))^2 / 6$
 $= 0.998188$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) = 2.756 \%$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 20.9$, $c = 3 \text{ mm}$, $E^* = 0.85$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 21$, $c = 3 \text{ mm}$, $E^* = 0.85$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -14 °C

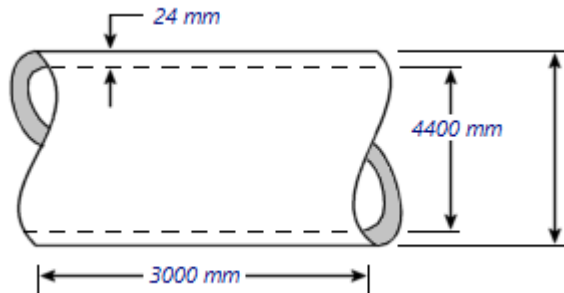
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.714	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 55.000 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.500 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-1**

Cylindrical Shell Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress	128.4 MPa	
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		41921412.0 mm²	SG of Contents	1.000	
Element Volume		45624.1 ltr	Weight of Contents	383456.9 N	
Empty Weight		76054.0 N	Operating Weight	459510.9 N	
			Comp. All. Ope. Stress	108.3 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03)$$

$$= 19.5332 + 3.0000 = 22.5332 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 21) / (2203 + 0.6 \cdot 21)$$

$$= 1.11 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (2200 + 0.6 \cdot 24)$$

$$= 1.27 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.03 \cdot (2203 + 0.6 \cdot 21)) / (0.85 \cdot 21)$$

$$= 128.410 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.542 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 24, tr = 21, c = 3 mm, E* = 0.85
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

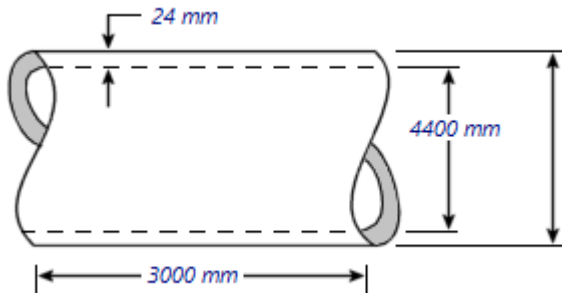
Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress		128.4 MPa
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		41921412.0 mm²	SG of Contents		1.000
Element Volume		45624.1 ltr	Weight of Contents		383456.9 N
Empty Weight		76054.0 N	Operating Weight		459510.9 N
			Comp. All. Ope. Stress		108.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03)$$

$$= 19.5332 + 3.0000 = 22.5332 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 21) / (2203 + 0.6 \cdot 21)$$

$$= 1.11 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (2200 + 0.6 \cdot 24)$$

$$= 1.27 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.03 \cdot (2203 + 0.6 \cdot 21)) / (0.85 \cdot 21)$$

$$= 128.410 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.542 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 24$, $t_r = 21$, $c = 3$ mm, $E^* = 0.85$

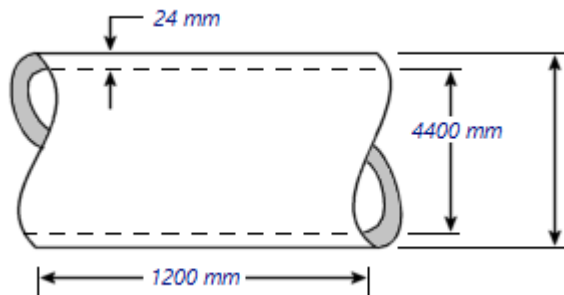
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -10 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-3**

Cylindrical Shell Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1200.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress		128.4 MPa
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		16768566.0 mm²	SG of Contents		1.000
Element Volume		18249.6 ltr	Weight of Contents		153382.8 N
Empty Weight		30421.7 N	Operating Weight		183804.5 N
			Comp. All. Ope. Stress		108.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03)$$

$$= 19.5332 + 3.0000 = 22.5332 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 21) / (2203 + 0.6 \cdot 21)$$

$$= 1.11 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (2200 + 0.6 \cdot 24)$$

$$= 1.27 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.03 \cdot (2203 + 0.6 \cdot 21)) / (0.85 \cdot 21)$$

$$= 128.410 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.542 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 24$, $t_r = 21$, $c = 3$ mm, $E^* = 0.85$

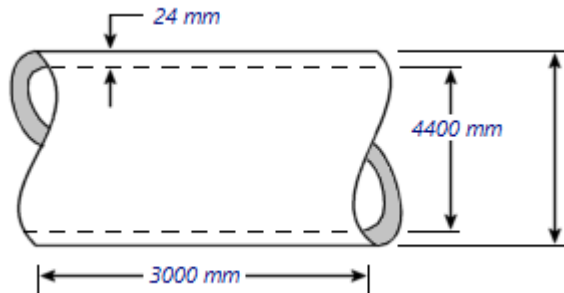
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -10 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-3**

Cylindrical Shell Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress	128.4 MPa	
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		41921412.0 mm²	SG of Contents	1.000	
Element Volume		45624.1 ltr	Weight of Contents	383456.9 N	
Empty Weight		76054.0 N	Operating Weight	459510.9 N	
			Comp. All. Ope. Stress	108.3 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03)$$

$$= 19.5332 + 3.0000 = 22.5332 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 21) / (2203 + 0.6 \cdot 21)$$

$$= 1.11 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (2200 + 0.6 \cdot 24)$$

$$= 1.27 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.03 \cdot (2203 + 0.6 \cdot 21)) / (0.85 \cdot 21)$$

$$= 128.410 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.542 %

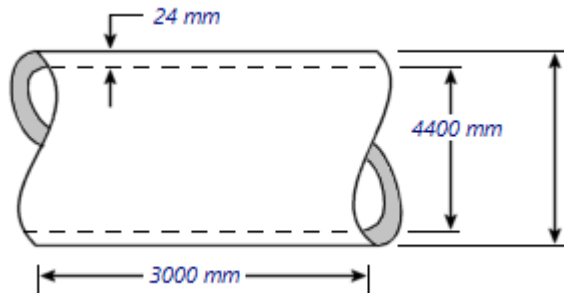
Minimum Design Metal Temperature Results:



Govrn. thk, tg = 24, tr = 21, c = 3 mm, E* = 0.85
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-4**

Cylindrical Shell Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress	128.4 MPa	
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		41921412.0 mm²	SG of Contents	1.000	
Element Volume		45624.1 ltr	Weight of Contents	383456.9 N	
Empty Weight		76054.0 N	Operating Weight	459510.9 N	
			Comp. All. Ope. Stress	108.3 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03)$$

$$= 19.5332 + 3.0000 = 22.5332 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 21) / (2203 + 0.6 \cdot 21)$$

$$= 1.11 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (2200 + 0.6 \cdot 24)$$

$$= 1.27 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.03 \cdot (2203 + 0.6 \cdot 21)) / (0.85 \cdot 21)$$

$$= 128.410 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.542 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 24$, $t_r = 21$, $c = 3$ mm, $E^* = 0.85$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -10 °C

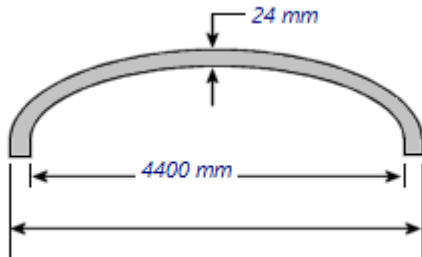
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-2

Elliptical Head Design Information:					
Design Pressure	P	1.035 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4400.000 mm	Actual Stress		127.6 MPa
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		22223490.0 mm²	SG of Contents		1.000
Element Volume		11913.0 ltr	Weight of Contents		104605.4 N
Empty Weight		51418.7 N	Operating Weight		156024.1 N
			Comp. All. Ope. Stress		108.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 55.000 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.03 \cdot 4406 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 1.03)$$

$$= 19.4118 + 3.0000 = 22.4118 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.035 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 21) / (1 \cdot 4406 + 0.2 \cdot 21)$$

$$= 1.12 - 0.035 = 1.08 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 24) / (1 \cdot 4400 + 0.2 \cdot 24)$$

$$= 1.28 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.03 \cdot (1 \cdot 4406 + 0.2 \cdot 21)) / (2 \cdot 0.85 \cdot 21)$$

$$= 127.570 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.03 \cdot 2203) / (138 \cdot 0.85 - 0.6 \cdot 1.03) + 3$
 $= 22.533 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.035 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 25) / (2203 + 0.6 \cdot 25)$
 $= 1.32 - 0.035 = 1.29 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (4406 / (2 \cdot 1103)))^2 / 6$
 $= 0.998188$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) = 2.756 \%$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 20.9$, $c = 3 \text{ mm}$, $E^* = 0.85$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 21$, $c = 3 \text{ mm}$, $E^* = 0.85$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -14 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.112 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.714	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 55.000 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.500 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.400 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.610 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.185 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.077 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 1.08 • 1

= 1.400 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	155.8	234.0	0.666	1.44
Shell-1	156.7	234.0	0.670	1.44
Shell-2	156.7	234.0	0.670	1.44
Shell-3	156.7	234.0	0.670	1.44
Shell-3	156.7	234.0	0.670	1.44
Shell-4	156.7	234.0	0.670	1.44
EH-2	155.8	234.0	0.666	1.44

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
K2A-2''	Nozzle	118.00	118.00	1.000
N3-6''	Nozzle	118.00	118.00	1.000
N3-6''	Pad	138.00	138.00	1.000
N4-4''	Nozzle	118.00	118.00	1.000
N4-4''	Pad	138.00	138.00	1.000
K2B-2''	Nozzle	118.00	118.00	1.000
K3-2''	Nozzle	118.00	118.00	1.000
K4-2''	Nozzle	118.00	118.00	1.000
N5-4''	Nozzle	118.00	118.00	1.000
N5-4''	Pad	138.00	138.00	1.000
N1-28''	Nozzle	118.00	118.00	1.000
N1-28''	Pad	138.00	138.00	1.000
N2-4''	Nozzle	118.00	118.00	1.000
N2-4''	Pad	138.00	138.00	1.000
N6-4''	Nozzle	118.00	118.00	1.000
N6-4''	Pad	138.00	138.00	1.000
K1A-2''	Nozzle	118.00	118.00	1.000
K1B-2''	Nozzle	118.00	118.00	1.000
M2-24''	Nozzle	118.00	118.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000



Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24''	36.09	234.00	0.154
K2A-2''	3.09	216.00	0.014
N3-6''	12.07	216.00	0.056
N4-4''	7.90	216.00	0.037
K2B-2''	3.09	216.00	0.014
K3-2''	3.09	216.00	0.014
K4-2''	3.09	216.00	0.014
N5-4''	7.90	216.00	0.037
N1-28''	25.09	216.00	0.116
N2-4''	5.53	216.00	0.026
N6-4''	6.84	216.00	0.032
K1A-2''	3.09	216.00	0.014
K1B-2''	3.09	216.00	0.014
M2-24''	36.09	216.00	0.167

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.400 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.610 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.185 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.077 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 1.08 • 1

= 1.400 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	178.0	234.0	0.761	1.44
Shell-1	179.2	234.0	0.766	1.44
Shell-2	179.2	234.0	0.766	1.44
Shell-3	179.2	234.0	0.766	1.44
Shell-3	179.2	234.0	0.766	1.44
Shell-4	179.2	234.0	0.766	1.44
EH-2	178.0	234.0	0.761	1.44

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
K2A-2''	Nozzle	118.00	118.00	1.000
N3-6''	Nozzle	118.00	118.00	1.000
N3-6''	Pad	138.00	138.00	1.000
N4-4''	Nozzle	118.00	118.00	1.000
N4-4''	Pad	138.00	138.00	1.000
K2B-2''	Nozzle	118.00	118.00	1.000
K3-2''	Nozzle	118.00	118.00	1.000
K4-2''	Nozzle	118.00	118.00	1.000
N5-4''	Nozzle	118.00	118.00	1.000
N5-4''	Pad	138.00	138.00	1.000
N1-28''	Nozzle	118.00	118.00	1.000
N1-28''	Pad	138.00	138.00	1.000
N2-4''	Nozzle	118.00	118.00	1.000
N2-4''	Pad	138.00	138.00	1.000
N6-4''	Nozzle	118.00	118.00	1.000
N6-4''	Pad	138.00	138.00	1.000
K1A-2''	Nozzle	118.00	118.00	1.000
K1B-2''	Nozzle	118.00	118.00	1.000
M2-24''	Nozzle	118.00	118.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000



Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24''	36.09	234.00	0.154
K2A-2''	3.09	216.00	0.014
N3-6''	12.07	216.00	0.056
N4-4''	7.90	216.00	0.037
K2B-2''	3.09	216.00	0.014
K3-2''	3.09	216.00	0.014
K4-2''	3.09	216.00	0.014
N5-4''	7.90	216.00	0.037
N1-28''	25.09	216.00	0.116
N2-4''	5.53	216.00	0.026
N6-4''	6.84	216.00	0.032
K1A-2''	3.09	216.00	0.014
K1B-2''	3.09	216.00	0.014
M2-24''	36.09	216.00	0.167

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****External Pressure Calculations:**

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	4448	21	0.00065573	65.5578
20	30	14033.3	4448	21	0.0001305	13.0469
30	40	14033.3	4448	21	0.0001305	13.0469
40	50	14033.3	4448	21	0.0001305	13.0469
50	60	14033.3	4448	21	0.0001305	13.0469
60	70	14033.3	4448	21	0.0001305	13.0469
70	80	No Calc	4448	21	0.00065573	65.5578

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	24	No Calc	...	0.34
20	30	24	No Calc	...	0.082
30	40	24	No Calc	...	0.082
40	50	24	No Calc	...	0.082
50	60	24	No Calc	...	0.082
60	70	24	No Calc	...	0.082
70	80	24	No Calc	...	0.34

Minimum

0.082

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	14033.3	No Calc	No Calc	No Calc
30	40	14033.3	No Calc	No Calc	No Calc
40	50	14033.3	No Calc	No Calc	No Calc
50	60	14033.3	No Calc	No Calc	No Calc
60	70	14033.3	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Element and Detail Weights:**

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	5243.6	11913	4681.78	11960.7	367.052
20	30	7755.87	45624.1	6790.96	45748.6	542.911
30	40	7755.87	45624.1	6790.96	45748.6	542.911
40	50	3102.36	18249.6	2716.4	18299.4	217.165
50	60	7755.87	45624.1	6790.96	45748.6	542.911
60	70	7755.87	45624.1	6790.96	45748.6	542.911
70	80	5243.6	11913	4681.78	11960.7	367.052
Total		44613	224571.92	39243	225215.36	3122

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	10667.5	-366.667	440	...	Liquid 10
10	Nozl	489.963	-1100	...	...	M1-24''
10	Wght	50	-1100	...	...	MH DAVIT-1
20	Sadl	1488.04	1150	2327	...	Left Saddle
20	Plat	11033.8	6650	2300.2	...	PF
20	Liqd	39104.3	1500	440	...	Liquid 20
20	Nozl	6.70272	500	2225.4	...	K2A-2''
20	Nozl	60.342	1500	2284.14	...	N3-6''
20	Nozl	31.2919	2900	2257.15	...	N4-4''
20	Nozl	6.70272	500	-2225.4	...	K2B-2''
20	Wght	130	1500	...	...	Cleat weight
20	Wght	735	1000	1905.26	1100	LL-1
30	Liqd	39104.3	1500	440	...	Liquid 30
30	Nozl	6.70272	1000	2225.4	...	K3-2''
30	Nozl	6.70272	2500	2225.4	...	K4-2''
40	Liqd	15641.7	600	440	...	Liquid 40
40	Nozl	31.2919	600	2257.15	...	N5-4''
40	Nozl	421.313	600	-2555.6	...	N1-28''
40	Nozl	19.0477	250	...	-2250.8	N2-4''
50	Liqd	39104.3	1500	440	...	Liquid 50
60	Sadl	1319.89	1850	2333	...	Left Saddle
60	Liqd	39104.3	1500	440	...	Liquid 60
60	Nozl	27.6286	1200	2257.15	...	N6-4''
60	Nozl	6.70272	2200	2225.4	...	K1A-2''
60	Nozl	6.68146	2200	-2225.4	...	K1B-2''
60	Wght	735	2000	1905.26	1100	LL-2
70	Liqd	10667.5	416.667	440	...	Liquid 70
70	Nozl	489.963	1150	...	...	M2-24''
70	Wght	50	1150	...	...	MH Davit-2

Total Weight of Each Detail Type:

Saddles	2807.9
Platforms	11033.8
Liquid	193394.0
Nozzles	1611.0
Weights	1700.0

Sum of the Detail Weights 210546.8 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	47735.9	47735.9	47735.9	47735.9	47735.9	47735.9



Saddles	2807.9	2807.9	2807.9	2807.9	2807.9	2807.9
Nozzles	1611.0	1611.0	1611.0	1611.0	1611.0	1611.0
Wld Weights	1700.0	1700.0	1700.0	1700.0	1700.0	1700.0
Platforms	...	...	...	11033.8	11033.8	11033.8
Ope. Liquid	...	...	...	...	...	193394.0
Test Liquid	...	224434.8	...	...	...	...
Totals	53854.9	278289.8	53854.9	64888.7	64888.7	258282.8

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	53854.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	278289.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	53854.9 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	64888.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	64888.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	258282.8 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	252537.7 kgm
Field Test Wt.	- Empty weight + test medium (full)	289323.5 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	22223490	21677106
20	30	41921412	41469020
30	40	41921412	41469020
40	50	16768566	16587610
50	60	41921412	41469020
60	70	41921412	41469020
70	80	22223490	21677106

Total	228901184.0	225817904.0 mm ²
Total	228.9	225.8 m ²

**Nozzle Flange MAWP Results: (MPa & °C)**

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M1-24''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K2A-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N3-6''	4.76	5.10	85	300	GR 1.1	3.509	3.636	4.763	3.636
N4-4''	4.76	5.10	85	300	GR 1.1	4.480	2.665	4.763	2.665
K2B-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K3-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K4-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N5-4''	4.76	5.10	85	300	GR 1.1	4.480	2.665	4.763	2.665
N1-28''	1.81	1.96	85	150	GR 1.1	1.416	1.224	1.812	1.302
N2-4''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N6-4''	4.76	5.10	85	300	GR 1.1	4.480	2.665	4.763	2.665
K1A-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K1B-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
M2-24''	1.81	1.96	85	150	GR 1.1	...	...	...	...
Min Rating	1.224	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3-6''	196.088	0.50	4725218	9600	1.077	3.636	Pass
N4-4''	138.176	0.50	2100096	6400	1.077	2.665	Pass
N5-4''	138.176	0.50	2100096	6400	1.077	2.665	Pass
N1-28''	736.600	0.46	102904760	44798	1.077	1.224	Pass
N6-4''	138.176	0.50	2100096	6400	1.077	2.665	Pass

Evaluating Flange on Nozzle [N3-6"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4725218) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 \cdot ((4.763 - 1.077) + .500 \cdot 4.763)$$

$$= 0.83136E+08 \leq 0.14377E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-6" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 83135992.0 / (\pi \cdot 196.088^3))$$

$$= \min(4.763, 3.636)$$

$$= 3.636 \text{ MPa}$$

Evaluating Flange on Nozzle [N4-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 \cdot ((4.763 - 1.077) + .500 \cdot 4.763)$$

$$= 0.37140E+08 \leq 0.50306E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 2.665)$$

$$= 2.665 \text{ MPa}$$



Evaluating Flange on Nozzle [N5-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(4.763 - 1.077) + .500 \cdot 4.763]$$

$$= 0.37140E+08 \leq 0.50306E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N5-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 2.665)$$

$$= 2.665 \text{ MPa}$$

Evaluating Flange on Nozzle [N1-28"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(102904760) + 4 \cdot \text{abs}(44798)736.600 \leq \pi \cdot 736.600^3 [(1.812 - 1.077) + .457 \cdot 1.812]$$

$$= 0.17785E+10 \leq 0.19634E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-28" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.812, 0.457 \cdot 1.812 + 1.812 - 1778522496.0 / (\pi \cdot 736.600^3))$$

$$= \min(1.812, 1.224)$$

$$= 1.224 \text{ MPa}$$

Evaluating Flange on Nozzle [N6-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(4.763 - 1.077) + .500 \cdot 4.763]$$

$$= 0.37140E+08 \leq 0.50306E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 2.665)$$

$$= 2.665 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Wind Load Calculation Results:****Input Values:**

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	0	
Base Elevation	0	mm
Percent Wind for Hydrottest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.738
Structure Height to Diameter ratio	3.298

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:

Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Z_g = 589788$ mm

Effective Height $[z]$:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 2500 + 0 = 2500 \text{ mm}$$

$$= 8.2 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z $[K_z]$:

$$\text{Because } z (8.2 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.41 (15 / Z_g)^{2/\alpha}$$

$$= 2.41 (15 / 1935)^{2/11.5}$$

$$= 1.04$$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:

$$= 1, \text{ per Table 26.6-1}$$

As there is No Hill Present: $[K_{zt}]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [1.59 \text{ kPa}]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.74 \cdot 45.1$$

$$= 939 \text{ lbs. } [4177 \text{ N}]$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
EH-1	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-1	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-2	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-3	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-3	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-4	2500.0	0.000	0.000	0.000	1.035	1.000	1.587
EH-2	2500.0	0.000	0.000	0.000	1.035	1.000	1.587

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PF	19931600.00	2500.00	1.59	31633.40	1.30

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
MH DAVIT-1	0.00	2500.00	1.59	0.00
Cleat weight	0.00	2500.00	1.59	0.00
LL-1	0.00	4405.26	1.64	0.00
LL-2	0.00	4405.26	1.64	0.00
MH Davit-2	0.00	2500.00	1.59	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	2500	5337.6	4193519	1.58714	4594.38
20	30	2500	5337.6	16012800	1.58714	52340.2
30	40	2500	5337.6	16012800	1.58714	17543.5
40	50	2500	5337.6	6405120	1.58714	7017.38
50	60	2500	5337.6	16012800	1.58714	17543.5
60	70	2500	5337.6	16012800	1.58714	17543.5
70	80	2500	5337.6	4193519	1.58714	4594.38

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(15.40.75)$
 = 0.156 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.16$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 2448849 \cdot 1.5$
 = 239497.453 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 2448849$
 = 106443.3 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	2200	272094	26610.8
20	30	2200	272094	26610.8
20	30	2200	272094	26610.8
30	40	2200	272094	26610.8
40	50	2200	272094	26610.8



50	60	2200	272094	26610.8
60	0	2200	272094	26610.8
60	70	2200	272094	26610.8
70	80	2200	272094	26610.8

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6323.624 mm
Center of Gravity of the Platforms	6700.000 mm
Center of Gravity of the Liquid	6650.000 mm
Center of Gravity of the Nozzles	6447.324 mm
Center of Gravity of the Added Weights (Operating)	6260.000 mm
Center of Gravity of the Added Weights (Empty)	6260.000 mm
Center of Gravity of Bare Shell New and Cold	6650.000 mm
Center of Gravity of Bare Shell Corroded	6650.000 mm
Vessel CG in the Operating Condition	6644.637 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	6614.610 mm
Vessel CG in the Test Condition	6643.151 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

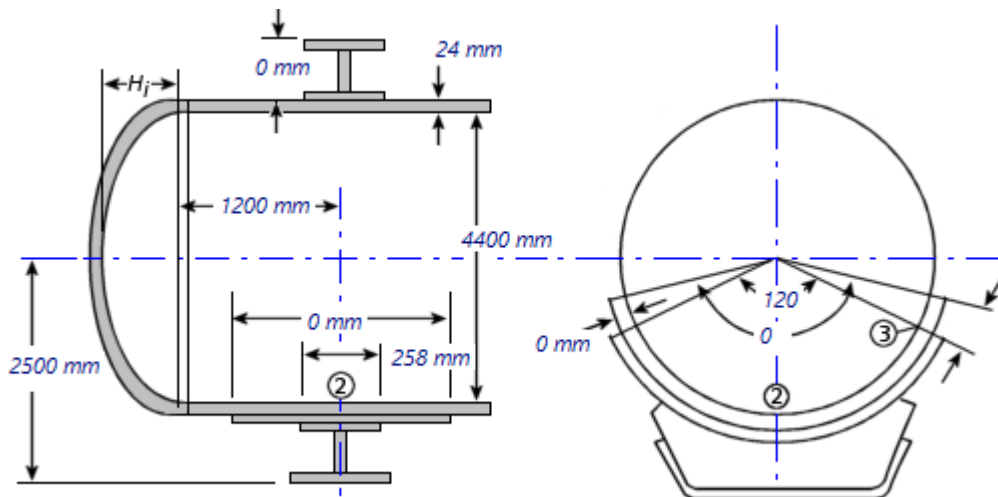
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{2214 \cdot 21}, 2 \cdot 1200)$$

$$= 594.3365 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	2213.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	13300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	258.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1113.50	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2500.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		1508918.00	N
Pressure used in Analysis	P	1.013	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	44.44	138.00
Long. Stress at Bottom of Midspan	62.32	138.00
Long. Stress at Top of Saddles	57.51	138.00
Long. Stress at Bottom of Saddles	51.10	138.00
Tangential Shear in Shell	28.02	110.40



Circ. Stress at Horn of Saddle	143.35	172.50
Circ. Compressive Stress in Shell	9.19	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (121177/2 + 0) \cdot 2500/3890$$

$$= 116815.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(49159, 0) \cdot 2500/10900$$

$$= 11275.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(239497, 0) \cdot 2500/10900$$

$$= 54930.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (239497/2 + 0) \cdot 2500/3890$$

$$= 230878.0 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1278039 + \max(11275, 116816, 54931, 230878)$$

$$= 1508918.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (5.78 + \pi/4 (4.45 \cdot 1.2)^2)$$

$$= 49161.652 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1523509.62	N
Transverse Shear Load	119748.73	N
Longitudinal Shear Load	239497.45	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3890.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0166	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1508918 \cdot 1200 [1 - (1 - 1200/13300 + (2214^2 - 1114^2)/(2 \cdot 1200 \cdot 13300))/(1 + (4 \cdot 1114)/(3 \cdot 13300))]$$

$$= -142098960.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1508918 \cdot 13300/4 ((1 + 2(2214^2 - 1114^2)/(13300^2))/(1 + (4 \cdot 1114)/(3 \cdot 13300)) - 4 \cdot 1200/13300)$$

$$= 2890557440.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 1.01 \cdot 2214/(2 \cdot 21) - 2.89E+09/(\pi \cdot 2214^2 \cdot 21)$$



$$= 44.44 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) + 2.89E+09 / (\pi \cdot 2214^2 \cdot 21)$$

$$= 62.32 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) - 142098960 / (0.11 \cdot \pi \cdot 2214^2 \cdot 21)$$

$$= 57.51 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) + 142098960 / (0.19 \cdot \pi \cdot 2214^2 \cdot 21)$$

$$= 51.10 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1508918 \cdot (13300 - 2 \cdot 1200) / (13300 + (4 \cdot 1114 / 3))$$

$$= 1112450.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1112450 / (2214 \cdot 21)$$

$$= 28.02 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2214 \cdot 21)}$$

$$= 168.168 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 1508918 \cdot 0.1 / (21(258 + 168 + 168))$$

$$= -9.19 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1508918 / (4 \cdot 21(258 + 168 + 168)) - 12 \cdot 0.017 \cdot 1508918 \cdot 2214 / (13300 \cdot 21^2)$$

$$= -143.35 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{(2214 \cdot 21)}, 2 \cdot 1200)$$

$$= 594.34 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 10900.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12005E-04 \cdot 10900(85 - 21.1)$$

$$= 8.360 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3890.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	280.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	251.0	mm
Friction Coefficient	mu	0.000	

**Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)**

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2203 \cdot 21)} + 330) 21^3) / 12$$

$$= 513628.344 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 513628 + 13976 (95.1 - 10.5)^2$$

$$= 100520704.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	665.5	21.0	10.5	13976.3	146751.1	0.101E+09
Wearplate	330.0	22.0	32.0	7260.0	232320.0	0.292E+08
Web	22.0	229.0	157.5	5038.0	793484.9	0.416E+08
BasePlate	280.0	25.0	284.5	7000.0	1991499.9	0.251E+09
Totals	...	...	...	33274.3	3164055.8	0.423E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3164055 / 33274$$

$$= 95.090 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1508918$$

$$= 307097.5312 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 15.9148 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2500 - 2203 \cdot \sin(120/2) / 1.05$$

$$= 678.134 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 208337648.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 46.8360 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(1508918 + 14592) 280 / (4 \cdot 3890 \cdot 161)}$$

$$= 22.580 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (2200 + 24 + 22)$$

$$= 3890.186 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 3890 / (4 - 1)$$



$$= 1296.729 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 1297 \cdot 280 / 2$$

$$= 181542.031 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 1523509 / (3890 \cdot 280)$$

$$= 1.399 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 181542 \cdot 1.4$$

$$= 253930.406 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 5192 + 14264$$

$$= 19456.014 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 253930 / 19456$$

$$= 13.053 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	258.0	...	5676.0	...	0.315E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 258 - 22$$

$$= 236.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 19456$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258 / 2$$

$$= 129.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{31484742 / 19456}$$

$$= 40.228 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2500 - \cos(120/2) (2200 + 24 + 22) - 25$$

$$= 1352.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (2200 + 24 + 22)$$

$$= 3890.186 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3890 - 50.8$$

$$= 3839.200 \text{ mm}$$



Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (3890 - 3839) / 2$$

$$= 25.493 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1352^2 + 25.5^2}$$

$$= 1352.240 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}{2 \cdot \pi^2 \cdot 199955 / 242}}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1352 / 40.2$$

$$= 33.615$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 239497 / (2 \cdot 3890) \cdot 1297 \cdot 1352 / 2$$

$$= 27000408.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($33.6 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (33.6)^2 / (2 \cdot 128^2)) 242 /$$

$$(5/3 + 3 \cdot (33.6) / (8 \cdot 128) - (33.6)^3 / (8 \cdot 128^3))$$

$$= 132.5 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 13.1 / 133 + (27000408 \cdot 129 / 31484772) / 161$$

$$= 0.784$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	236.0	0.0	5192.0	0.0	0.315E+08
Web	1296.7	22.0	0.0	28528.0	0.0	0.115E+07
Totals	...	...	...	33720.0	...	0.326E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 33720$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258 / 2$$

$$= 129.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 2500 - \sqrt{(2246 + 22)^2 - (1297/2)^2} - 25$$

$$= 324.619 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{32615852 / 33720}$$

$$= 31.101 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 325 / 31.1$$



$$= 10.438$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 239497 / (2 \cdot 3890)$$

$$= 30.784 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 30.8 \cdot 1297 \cdot 325$$

$$= 12963435.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 12963435 / 252836$$

$$= 51.256 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($10.4 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.4)^2 / (2 \cdot 128^2)) 242 /$$

$$(5/3 + 3 \cdot (10.4) / (8 \cdot 128) - (10.4)^3 / (8 \cdot 128^3))$$

$$= 142.1 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 15.1 / 142 + (12963435 \cdot 129 / 32615852) / 161$$

$$= 0.424$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.90 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	981.5097 mm ²
Saddle Load QO (Weight)	QO	1292631.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	54930.6 N
Maximum Transverse Force	Ft	119748.7 N
Maximum Longitudinal Force	Fl	239497.5 N
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

$$= 387058.6 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 387059 / (982 \cdot 4)$$

$$= 98.6 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 119749 / (982 \cdot 4)$$

$$= 30.5 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 239497 / (99.9 \cdot 4)$$

$$= 599.3939 \text{ mm}^2$$

**Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2500 \cdot 119749 + 0$$

$$= 299493216.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 299493216 / 1292631$$

$$= 231.60 \text{ mm} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 599, 0]$$

$$= 599.3939 \text{ mm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 22, c = 3 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.85, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -13 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{2214 \cdot 21}, 2 \cdot 1200)$$

$$= 594.3365 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	2213.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	13300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	258.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1113.50	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2500.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		1509359.38	N
Pressure used in Analysis	P	1.013	MPa

Horizontal Vessel Analysis Results:	Actual	Allowable
-------------------------------------	--------	-----------



	MPa	MPa
Long. Stress at Top of Midspan	44.44	138.00
Long. Stress at Bottom of Midspan	62.33	138.00
Long. Stress at Top of Saddles	57.51	138.00
Long. Stress at Bottom of Saddles	51.10	138.00
Tangential Shear in Shell	28.03	110.40
Circ. Stress at Horn of Saddle	143.39	172.50
Circ. Compressive Stress in Shell	9.19	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (121177/2 + 0) \cdot 2500/3869$$

$$= 117437.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(49159, 0) \cdot 2500/10900$$

$$= 11275.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(239497, 0) \cdot 2500/10900$$

$$= 54930.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (239497/2 + 0) \cdot 2500/3869$$

$$= 232107.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1277252 + \max(11275, 117437, 54931, 232107)$$

$$= 1509359.4 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (5.78 + \pi/4(4.45 \cdot 1.2)^2)$$

$$= 49161.652 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1522302.25	N
Transverse Shear Load	119748.73	N
Longitudinal Shear Load	239497.45	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(4400/2 + 24 + 10)\sin(120/2), 2 \cdot 2214)$$

$$= 3869.402 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0166	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -1509359 \cdot 1200 [1 - (1 - 1200/13300 + (2214^2 - 1114^2)/(2 \cdot 1200 \cdot 13300))/(1 + (4 \cdot 1114)/(3 \cdot 13300))]$$

$$= -142140544.0 \text{ N-mm}$$



Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 1509359 \cdot 13300 / 4 \cdot ((1 + 2(2214^2 - 1114^2) / (13300^2)) / (1 + (4 \cdot 1114) / (3 \cdot 13300)) - 4 \cdot 1200 / 13300)$$

$$= 2891403520.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) - 2.89E+09 / (\pi \cdot 2214^2 \cdot 21)$$

$$= 44.44 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) + 2.89E+09 / (\pi \cdot 2214^2 \cdot 21)$$

$$= 62.33 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) - 142140544 / (0.11 \cdot \pi \cdot 2214^2 \cdot 21)$$

$$= 57.51 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1.01 \cdot 2214 / (2 \cdot 21) + 142140544 / (0.19 \cdot \pi \cdot 2214^2 \cdot 21)$$

$$= 51.10 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1509359 \cdot (13300 - 2 \cdot 1200) / (13300 + (4 \cdot 1114 / 3))$$

$$= 1112775.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 1112775 / (2214 \cdot 21)$$

$$= 28.03 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2214 \cdot 21)}$$

$$= 168.168 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1509359 \cdot 0.1 / (21(258 + 168 + 168))$$

$$= -9.19 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -1509359 / (4 \cdot 21(258 + 168 + 168)) - 12 \cdot 0.017 \cdot 1509359 \cdot 2214 / (13300 \cdot 21^2)$$

$$= -143.39 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{(2214 \cdot 21)}, 2 \cdot 1200)$$

$$= 594.34 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3890.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	280.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	251.0	mm



$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 3869 / (4 - 1) \\
 &= 1289.800 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 1290 \cdot 280 / 2 \\
 &= 180572.078 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 1522302 / (3890 \cdot 280) \\
 &= 1.398 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 180572 \cdot 1.4 \\
 &= 252373.547 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 5192 + 14188 \\
 &= 19379.805 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 252374 / 19380 \\
 &= 13.024 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	258.0	...	5676.0	...	0.315E+08

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 258 - 22 \\
 &= 236.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 19380 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 258 / 2 \\
 &= 129.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 31484742 / 19380 } \\
 &= 40.307 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2500 - \cos(120 / 2) (2200 + 24 + 10) - 25 \\
 &= 1358.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199955 / 242 } \\
 &= 127.710
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$= \text{KL} / \text{r}$$



$$= 1 \cdot 1358/40.3$$

$$= 33.692$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bpln) \cdot e \cdot L / 2$$

$$= 239497 / (2 \cdot 3890) \cdot 1290 \cdot 1358/2$$

$$= 26970538.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($33.7 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (33.7)^2 / (2 \cdot 128^2)) 242 /$$

$$(5/3 + 3 \cdot (33.7) / (8 \cdot 128) - (33.7^3) / (8 \cdot 128^3))$$

$$= 132.5 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 13/132 + (26970538 \cdot 129/31484772) / 161$$

$$= 0.783$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	236.0	0.0	5192.0	0.0	0.315E+08
Web	1289.8	22.0	0.0	28375.6	0.0	0.114E+07
Totals	...	...	...	33567.6	...	0.326E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/33568$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258/2$$

$$= 129.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 2500 - \sqrt{ (2234 + 10)^2 - (1290/2)^2 } - 25$$

$$= 336.108 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 32609704/33568 }$$

$$= 31.168 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 336/31.2$$

$$= 10.784$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 239497 / (2 \cdot 3890)$$

$$= 30.784 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 30.8 \cdot 1290 \cdot 336$$

$$= 13350528.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13350528/252789$$

$$= 52.796 \text{ MPa}$$

Compressive Allowable, $KL/r < Cc$ ($10.8 < 128$) per AISC E2-1 [Sca]:



$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (10.8)^2 / (2 \cdot 128^2)) 242 / (5/3 + 3 \cdot (10.8) / (8 \cdot 128) - (10.8^3) / (8 \cdot 128^3))$$

$$= 142 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 15.1/142 + (13350528 \cdot 129/32609704) / 161$$

$$= 0.433$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.90 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	981.5097 mm ²
Saddle Load QO (Weight)	QO	1290195.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	54930.6 N
Maximum Transverse Force	Ft	119748.7 N
Maximum Longitudinal Force	Fl	239497.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 239497 / (982 \cdot 4)$$

$$= 61.0 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 119749 / (982 \cdot 4)$$

$$= 30.5 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 239497 / (99.9 \cdot 4)$$

$$= 599.3939 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2500 \cdot 119749 + 0$$

$$= 299493216.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 299493216 / 1290195$$

$$= 232.04 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0, 599, 0]$$

$$= 599.3939 \text{ mm}^2$$



MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 0.85$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

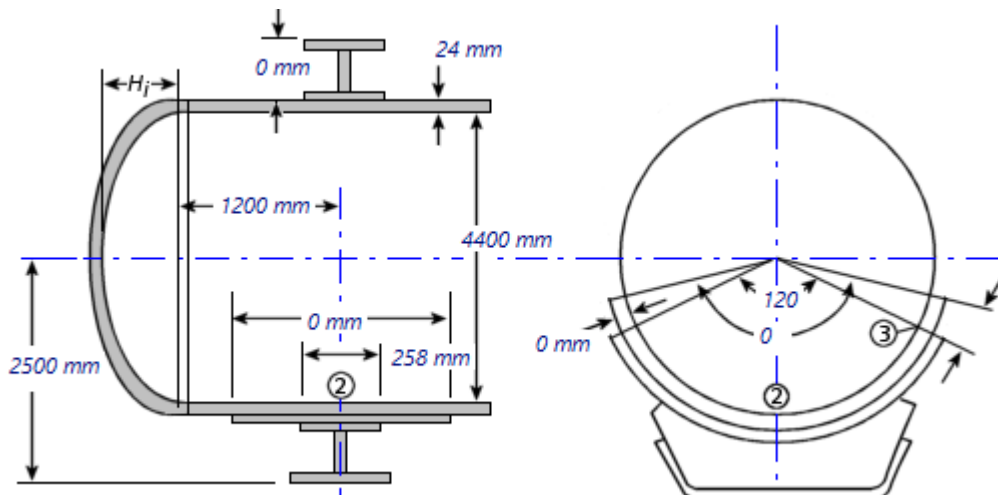
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{2212 \cdot 24}, 2 \cdot 1200)$$

$$= 617.4370 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	R_m	2212.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	13300.00	mm
Distance from Saddle to Vessel tangent	a_1 or a	1200.00	mm
Saddle Width	b_1 or b	258.00	mm
Saddle Bearing Angle	δ or θ	120.00	degrees
Inside or Mean Depth of Head	H_i or h_m	1112.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2500.00	mm
Coefficient of Friction	μ	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1443728.38	N
Pressure used in Analysis	P	1.422 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	58.04	234.00
Long. Stress at Bottom of Midspan	73.03	234.00
Long. Stress at Top of Saddles	68.99	234.00
Long. Stress at Bottom of Saddles	63.62	234.00
Tangential Shear in Shell	23.48	138.58



Circ. Stress at Horn of Saddle	107.32	350.99
Circ. Compressive Stress in Shell	7.41	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (39988/2 + 0) \cdot 2500/3890$$

$$= 38549.1 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(16223, 0) \cdot 2500/10900$$

$$= 3720.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1405179 + \max(3721, 38549, 0, 0)$$

$$= 1443728.4 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (5.78 + \pi/4 (4.45 \cdot 1.2)^2)$$

$$= 49161.652 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1458320.00	N
Transverse Shear Load	19994.16	N
Longitudinal Shear Load	16222.51	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3890.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0166	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -1443728 \cdot 1200 [1 - (1 - 1200/13300 + (2212^2 - 1112^2) / (2 \cdot 1200 \cdot 13300)) / (1 + (4 \cdot 1112) / (3 \cdot 13300))]$$

$$= -135905232.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/3L) - 4a/L)$$

$$= 1443728 \cdot 13300 / 4 ((1 + 2(2212^2 - 1112^2)/(13300^2)) / (1 + (4 \cdot 1112) / (3 \cdot 13300)) - 4 \cdot 1200/13300)$$

$$= 2766124800.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212 / (2 \cdot 24) - 2.77E+09 / (\pi \cdot 2212^2 \cdot 24)$$

$$= 58.04 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212 / (2 \cdot 24) + 2.77E+09 / (\pi \cdot 2212^2 \cdot 24)$$

$$= 73.03 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212 / (2 \cdot 24) - 135905232 / (0.11 \cdot \pi \cdot 2212^2 \cdot 24)$$



= 68.99 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212 / (2 \cdot 24) + 135905232 / (0.19 \cdot \pi \cdot 2212^2 \cdot 24)$$

$$= 63.62 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1443728 (13300 - 2 \cdot 1200) / (13300 + (4 \cdot 1112 / 3))$$

$$= 1064533.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1064533 / (2212 \cdot 24)$$

$$= 23.48 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{2212 \cdot 24}$$

$$= 179.718 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1443728 \cdot 0.1 / (24(258 + 180 + 180))$$

$$= -7.41 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1443728 / (4 \cdot 24(258 + 180 + 180)) - 12 \cdot 0.017 \cdot 1443728 \cdot 2212 / (13300 \cdot 24^2)$$

$$= -107.32 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{2212 \cdot 24}, 2 \cdot 1200)$$

$$= 617.44 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3890.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	280.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	251.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{2200 \cdot 24} + 330)^3) / 12$$

$$= 793105.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 793106 + 16523(91.4 - 12)^2$$

$$= 104918656.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	688.5	24.0	12.0	16523.1	198276.7	0.105E+09
Wearplate	330.0	22.0	35.0	7260.0	254100.0	0.234E+08
Web	22.0	229.0	160.5	5038.0	808598.9	0.461E+08
BasePlate	280.0	25.0	287.5	7000.0	2012500.0	0.270E+09



Totals	...	...	...	35821.1	3273475.8	0.444E+09
--------	-----	-----	-----	---------	-----------	-----------

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3273475/35821$$

$$= 91.384 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1443728$$

$$= 293830.0000 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 15.2272 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2500 - 2200 \cdot \sin(120/2) / 1.05$$

$$= 680.615 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 200066096.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 41.1672 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (1443728 + 14592) \cdot 280) / (4 \cdot 3890 \cdot 173)}$$

$$= 21.313 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (2200 + 24 + 22)$$

$$= 3890.186 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 3890 / (4 - 1)$$

$$= 1296.729 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1297 \cdot 280 / 2$$

$$= 181542.031 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1458320 / (3890 \cdot 280)$$

$$= 1.339 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 181542 \cdot 1.34$$

$$= 243064.938 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



$$= 5192 + 14264$$

$$= 19456.014 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 243065/19456$$

$$= 12.494 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	258.0	...	5676.0	...	0.315E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 258 - 22$$

$$= 236.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/19456$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258/2$$

$$= 129.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{31484742/19456}$$

$$= 40.228 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2500 - \cos(120/2) (2200 + 24 + 22) - 25$$

$$= 1352.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (2200 + 24 + 22)$$

$$= 3890.186 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3890 - 50.8$$

$$= 3839.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (3890 - 3839) / 2$$

$$= 25.493 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1352^2 + 25.5^2}$$

$$= 1352.240 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955/260}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1352/40.2$$



$$= 33.615$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 16223 / (2 \cdot 3890) \cdot 1297 \cdot 1352 / 2$$

$$= 1828889.375 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (33.6 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (33.6)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (33.6) / (8 \cdot 123) - (33.6^3) / (8 \cdot 123^3))$$

$$= 141.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.5 / 142 + (1828889 \cdot 129 / 31484772) / 173$$

$$= 0.131$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	236.0	0.0	5192.0	0.0	0.315E+08
Web	1296.7	22.0	0.0	28528.0	0.0	0.115E+07
Totals	...	...	...	33720.0	...	0.326E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 33720$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258 / 2$$

$$= 129.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 2500 - \sqrt{ (2246 + 22)^2 - (1297/2)^2 } - 25$$

$$= 324.619 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (32615852 / 33720) }$$

$$= 31.101 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 325 / 31.1$$

$$= 10.438$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 16223 / (2 \cdot 3890)$$

$$= 2.085 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.09 \cdot 1297 \cdot 325$$

$$= 878086.250 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 878086 / 252836$$

$$= 3.472 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (10.4 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



$$= (1 - (10.4)^2 / (2 \cdot 123^2)) \cdot 260 / (5/3 + 3 \cdot (10.4) / (8 \cdot 123) - (10.4^3) / (8 \cdot 123^3))$$

$$= 152.5 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 14.4/153 + (878086 \cdot 129/32615852) / 173$$

$$= 0.115$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	981.5097 mm ²
Saddle Load QO (Weight)	QO	1419770.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	3720.8 N
Maximum Transverse Force	Ft	19994.2 N
Maximum Longitudinal Force	Fl	16222.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (Bolt \text{ Area} \cdot \text{Number of Bolts})$$

$$= 16223 / (982 \cdot 4)$$

$$= 4.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (Bolt \text{ Area} \cdot \text{Number of Bolts})$$

$$= 19994 / (982 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. MossBolt Area Requirement Due to Longitudinal Load [Bltarear_l]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarear_s]:

$$= Fl / (Bolt \text{ Shear Allowable} \cdot \text{Nbolts})$$

$$= 16223 / (68.4 \cdot 4)$$

$$= 59.2989 \text{ mm}^2$$

Bolt Area due to Transverse Load:Moment on Baseplate Due to Transverse Load [R_{mom}]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2500 \cdot 19994 + 0$$

$$= 50005660.00 \text{ N-mm}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 50005660 / 1419770$$

$$= 35.21 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltarear_t]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarear_l, Bltarear_s, Bltarear_t]$$

$$= \max[0, 59.3, 0]$$

$$= 59.2989 \text{ mm}^2$$

**Note:**

Wear Pad Width (330.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{ 2212 \cdot 24 }, 2 \cdot 1200)$$

$$= 617.4370 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	2212.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	13300.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	258.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1112.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2500.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1443146.25	N
Pressure used in Analysis	P	1.422	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	58.04	234.00
Long. Stress at Bottom of Midspan	73.03	234.00
Long. Stress at Top of Saddles	68.99	234.00
Long. Stress at Bottom of Saddles	63.62	234.00
Tangential Shear in Shell	23.47	138.58
Circ. Stress at Horn of Saddle	107.27	350.99
Circ. Compressive Stress in Shell	7.40	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (39988/2 + 0) \cdot 2500/3869$$

$$= 38754.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(16223, 0) \cdot 2500/10900$$

$$= 3720.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1404391 + \max(3721, 38754, 0, 0)$$

$$= 1443146.2 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (5.78 + \pi/4 (4.45 \cdot 1.2)^2)$$

$$= 49161.652 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

Vertical Load (including saddle weight)	1456089.12	N
Transverse Shear Load	19994.16	N
Longitudinal Shear Load	16222.51	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(4400/2 + 24 + 10) \sin(120/2), 2 \cdot 2212)$$

$$= 3869.402 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0166	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -1443146 \cdot 1200 [1 - (1 - 1200/13300 + (2212^2 - 1112^2)/(2 \cdot 1200 \cdot 13300))/(1 + (4 \cdot 1112)/(3 \cdot 13300))]$$

$$= -135850432.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h m^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 1443146 \cdot 13300/4((1 + 2(2212^2 - 1112^2)/(13300^2))/(1 + (4 \cdot 1112)/(3 \cdot 13300)) - 4 \cdot 1200/13300)$$

$$= 2765009408.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 1.42 \cdot 2212/(2 \cdot 24) - 2.77E+09/(\pi \cdot 2212^2 \cdot 24)$$

$$= 58.04 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212/(2 \cdot 24) + 2.77E+09/(\pi \cdot 2212^2 \cdot 24)$$

$$= 73.03 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1.42 \cdot 2212/(2 \cdot 24) - 135850432/(0.11 \cdot \pi \cdot 2212^2 \cdot 24)$$

$$= 68.99 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.42 \cdot 2212/(2 \cdot 24) + 135850432/(0.19 \cdot \pi \cdot 2212^2 \cdot 24)$$

$$= 63.62 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 1443146(13300 - 2 \cdot 1200)/(13300 + (4 \cdot 1112/3))$$

$$= 1064104.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1064104/(2212 \cdot 24)$$

$$= 23.47 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2212 \cdot 24)}$$

$$= 179.718 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$



$$= -0.76 \cdot 1443146 \cdot 0.1 / (24 (258 + 180 + 180))$$

$$= -7.40 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) $[\sigma_7^*]$:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1443146 / (4 \cdot 24 (258 + 180 + 180)) -$$

$$12 \cdot 0.017 \cdot 1443146 \cdot 2212 / (13300 \cdot 24^2)$$

$$= -107.27 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(258 + 1.56 \cdot \sqrt{2212 \cdot 24}, 2 \cdot 1200)$$

$$= 617.44 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3890.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	280.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	251.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{2200 \cdot 24} + 330) 24^3) / 12$$

$$= 793105.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 793106 + 16523 (97.3 - 12)^2$$

$$= 121004680.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	688.5	24.0	12.0	16523.1	198276.7	0.121E+09
Wearplate	330.0	10.0	29.0	3300.0	95700.0	0.154E+08
Web	22.0	241.0	154.5	5302.0	819159.1	0.430E+08
BasePlate	280.0	25.0	287.5	7000.0	2012500.4	0.254E+09
Totals	...	...	...	32125.1	3125636.2	0.433E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3125636 / 32125$$

$$= 97.296 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 1443146$$

$$= 293711.5625 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / A_s) = 18.8269 \text{ MPa}$$



Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$

Saddle Splitting Dimension [d]:

$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 2500 - 2200 \cdot \sin(120/2) / 1.05$
 $= 680.615 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 199985440.0000 \text{ N-mm}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 44.9181 \text{ MPa}$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(1443146 + 12943)280 / (4 \cdot 3890 \cdot 173)}$
 $= 21.297 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (2200 + 24 + 10)$
 $= 3869.401 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 3869 / (4 - 1)$
 $= 1289.800 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$
 $= 1290 \cdot 280 / 2$
 $= 180572.078 \text{ mm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 1456089 / (3890 \cdot 280)$
 $= 1.337 \text{ N/mm}^2$

Axial Load [P]:

$= A_p \cdot B_p$
 $= 180572 \cdot 1.34$
 $= 241396.438 \text{ N}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$
 $= 5192 + 14188$
 $= 19379.805 \text{ mm}^2$

Compressive Stress [Sc]:

$= P / A_r$
 $= 241396 / 19380$
 $= 12.457 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	258.0	...	5676.0	...	0.315E+08

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$
 $= 258 - 22$
 $= 236.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$



$$= 0/19380$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258/2$$

$$= 129.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{31484742/19380}$$

$$= 40.307 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2500 - \cos(120/2) (2200 + 24 + 10) - 25$$

$$= 1358.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1358/40.3$$

$$= 33.692$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 16223 / (2 \cdot 3890) \cdot 1290 \cdot 1358/2$$

$$= 1826866.125 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($33.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (33.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (33.7) / (8 \cdot 123) - (33.7^3) / (8 \cdot 123^3))$$

$$= 141.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 12.5/142 + (1826866 \cdot 129/31484772) / 173$$

$$= 0.131$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	236.0	0.0	5192.0	0.0	0.315E+08
Web	1289.8	22.0	0.0	28375.6	0.0	0.114E+07
Totals	...	...	...	33567.6	...	0.326E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/33568$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 258/2$$

$$= 129.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2500 - \sqrt{(2234 + 10)^2 - (1290/2)^2} - 25$$

$$= 336.108 \text{ mm}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{32609704/33568}$$

$$= 31.168 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 336/31.2$$

$$= 10.784$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 16223 / (2 \cdot 3890)$$

$$= 2.085 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.09 \cdot 1290 \cdot 336$$

$$= 904306.250 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 904306/252789$$

$$= 3.576 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($10.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (10.8) / (8 \cdot 123) - (10.8)^3 / (8 \cdot 123^3))$$

$$= 152.4 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 14.4/152 + (904306 \cdot 129/32609704)/173$$

$$= 0.115$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	48.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	981.5097 mm ²
Saddle Load QO (Weight)	QO	1417334.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	3720.8 N
Maximum Transverse Force	Ft	19994.2 N
Maximum Longitudinal Force	Fl	16222.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 16223 / (982 \cdot 4)$$

$$= 4.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19994 / (982 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:



= $F / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $16223 / (68.4 \cdot 4)$
= 59.2989 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$
= $2500 \cdot 19994 + 0$
= 50005660.00 N-mm

Eccentricity (e):

= $Rmom / QO$
= $50005660 / 1417334$
= 35.27 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 59.3, 0]$
= 59.2989 mm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

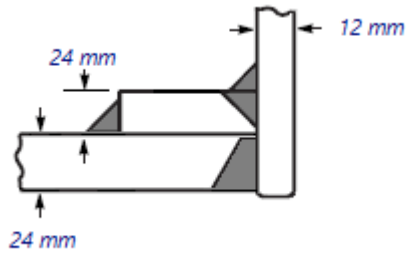
**Input, Nozzle Desc: M1-24''****From: 10**

Pressure for Reinforcement Calculations	P	1.0902	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	4400.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.09 \cdot 4406 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.09)$
 $= 15.6554 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.09 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1.09)$
 $= 2.4003 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001	mm
Parallel to Vessel Wall, opening length	d	591.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.5000	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

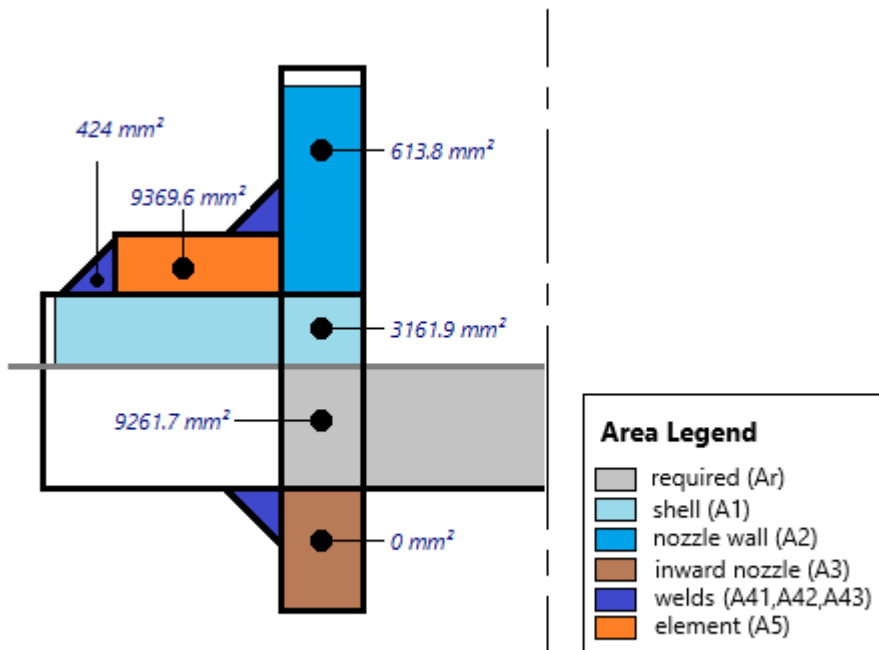


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9261.726	NA	NA
Area in Shell	A1	3161.875	NA	NA
Area in Nozzle Wall	A2	613.768	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	424.000	NA	NA
Area in Element	A5	9369.601	NA	NA
TOTAL AREA AVAILABLE	Atot	13569.244	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	820.5201	24.0000 mm
Based on given Pad Diameter:	1000.0000	12.9664 mm
Based on Shell or Nozzle Thickness:	1031.4402	12.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (592 \cdot 15.7 \cdot 1 + 2 \cdot 9 \cdot 15.7 \cdot 1 \cdot (1 - 1))$$

$$= 9261.726 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 592 (1 \cdot 21 - 1 \cdot 15.7) - 2 \cdot 9 (1 \cdot 21 - 1 \cdot 15.7) \cdot (1 - 1)$$

$$= 3161.875 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 46.5) \cdot (9 - 2.4) \cdot 1$$

$$= 613.768 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 18^2 \cdot 1$$

$$= 424.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (1000 - 610) \cdot 24 \cdot 1$$

$$= 9369.601 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.4003 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.4003 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 12$, $t_r = 2.4$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.4$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.4$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 24$, $t_r = 15.7$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.4$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.09/1.96 = 0.556

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (9262 - 3162 + 2 \cdot 9 \cdot 1 \cdot (1 \cdot 21 - 15.7)) \cdot 138)$
 $= 854967.19 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (614 + 9370 + 424 - 0 \cdot 1) \cdot 138$
 $= 1436068.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (614 + 0 + 100 + (378)) \cdot 138$
 $= 150648.50 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (614 + 0 + 424 + 9370 + (378)) \cdot 138$
 $= 1488227.38 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 647434. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1000 \cdot 18 \cdot 0.49 \cdot 138$
 $= 1911713. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 300) \cdot (12 - 3) \cdot 0.7 \cdot 138$
 $= 820125. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 24 \cdot 0.74 \cdot 138$
 $= 2346616. \text{ N}$



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (24 - 3) \cdot 0.74 \cdot 138$
 $= 2053289. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (1911713 + 820125) = 2731838 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (647434 + 2346616 + 2053289 + 0) = 5047339 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (1911713 + 2053289 + 0) = 3965002 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2731838 N	, must exceed W = 854967 N	or W1 = 1436068 N
Path 2-2 = 5047339 N	, must exceed W = 854967 N	or W2 = 150648 N
Path 3-3 = 3965002 N	, must exceed W = 854967 N	or W3 = 1488227 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.6908 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.6909 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K2A-2"

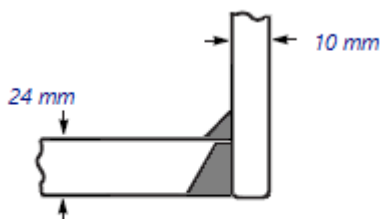
From: 20

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
= 17.2781 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.08 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.08)$
= 0.2310 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.8000	mm
Parallel to Vessel Wall	Rn+tn+t	46.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2310	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 20.2781	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.2781	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(20.3, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.23, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.08 / 1.96 = 0.550$

Weld Size Calculations, Description: K2A-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
---	------	--------	----



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1467 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N3-6''

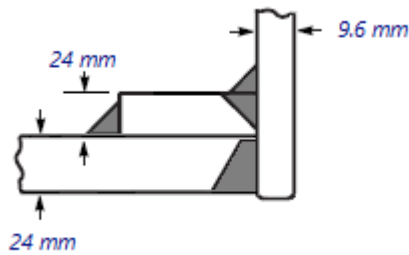
From: 20

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		65.8625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
 $= 17.2781 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.08 \cdot 84.1 / (118 \cdot 1 + 0.4 \cdot 1.08)$
 $= 0.7653 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452 mm
Parallel to Vessel Wall, opening length	d	155.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		40.5030 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

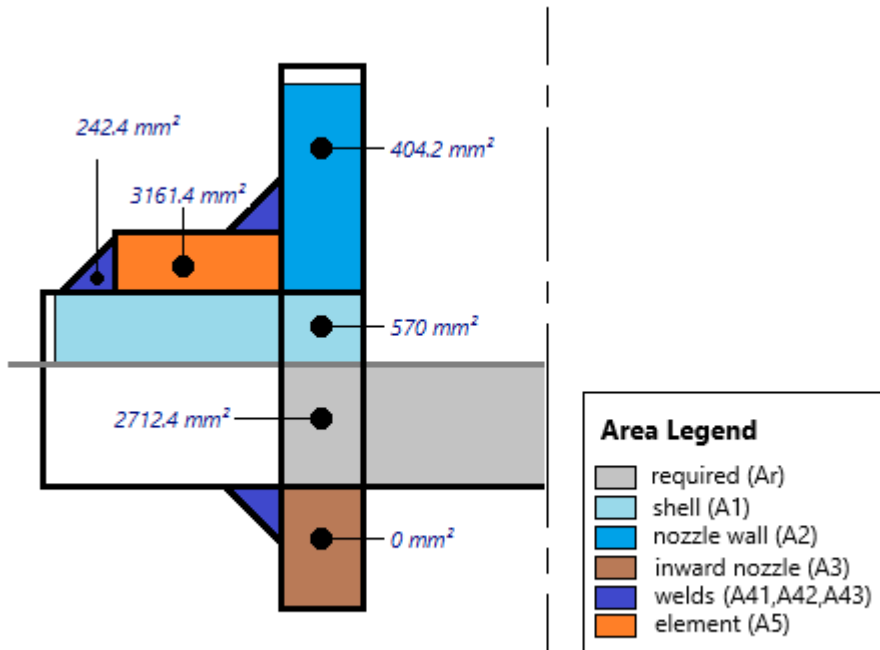


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2712.420	NA	NA
Area in Shell	A1	570.043	NA	NA
Area in Nozzle Wall	A2	404.228	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	242.389	NA	NA
Area in Element	A5	3161.400	NA	NA
TOTAL AREA AVAILABLE	Atot	4378.060	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	230.5984	24.0000 mm
Based on given Pad Diameter:	300.0000	11.3552 mm
Based on the Estimated Diameter Limit:	308.5577	10.6625 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155 \cdot 17.3 \cdot 1 + 2 \cdot 6.6 \cdot 17.3 \cdot 1 \cdot (1 - 0.86))$
 $= 2712.420 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 155(1 \cdot 21 - 1 \cdot 17.3) - 2 \cdot 6.6$
 $(1 \cdot 21 - 1 \cdot 17.3) \cdot (1 - 0.86)$
 $= 570.043 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 40.5) \cdot (6.6 - 0.77) \cdot 0.86$$

$$= 404.228 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100) \cdot 0.86 + (0) \cdot 0.86 + 157^2 \cdot 1$$

$$= 242.389 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr4}$$

$$= (300 - 168) \cdot 24 \cdot 1$$

$$= 3161.400 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.7653 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 20.2781 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 20.2781 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 9.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[9.22, \max(20.3, 4.5)]$$

$$= 9.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.77, 9.22)$$

$$= 9.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	255.6 MPa	Passed
Occasional	:	6.1,	Allowable	:	156.9 MPa	Passed
Shear	:	24.8,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.6, t_r = 0.77, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.12, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.6, t_r = 0.77, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.12, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.6, t_r = 0.77, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.12, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 17.3, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.6, tr = 0.77, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.08/5.10 = 0.211$

Weld Size Calculations, Description: N3-6"

Intermediate Calc. for nozzle/shell Welds Tmin 6.6012 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.6208 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (2712 - 570 + 2 \cdot 6.6 \cdot 0.86 \cdot (1 \cdot 21 - 17.3)) \cdot 138)$$

$$= 301415.31 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (404 + 3161 + 242 - 0 \cdot 0.86) \cdot 138$$

$$= 525452.12 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (404 + 0 + 85.5 + (237)) \cdot 138$$

$$= 100288.62 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (404 + 0 + 242 + 3161 + (237)) \cdot 138$$

$$= 558164.31 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118 \\
 &= 152817. \text{ N}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 300 \cdot 18 \cdot 0.49 \cdot 138 \\
 &= 573514. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 80.8) \cdot (9.6 - 3) \cdot 0.7 \cdot 118 \\
 &= 138458. \text{ N}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 168 \cdot 22 \cdot 0.74 \cdot 138 \\
 &= 593784. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 168 \cdot (24 - 3) \cdot 0.74 \cdot 138 \\
 &= 566793. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (573514 + 138458) = 711972 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (152817 + 593784 + 566793 + 0) = 1313394 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (573514 + 566793 + 0) = 1140307 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 711971 N	, must exceed W = 301415 N	or W1 = 525452 N
Path 2-2 = 1313394 N	, must exceed W = 301415 N	or W2 = 100288 N
Path 3-3 = 1140307 N	, must exceed W = 301415 N	or W3 = 558164 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.6094 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.6095 mm

Input Echo, WRC297 Item 1, Description: N3-6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4400.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	168.275	mm
Nozzle Thickness	9.6012	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	300.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	1.08	MPa
Radial Load	P	-9599.58	N
Circumferential Shear	Vc	7199.67	N
Longitudinal Shear	Vl	9599.58	N
Circumferential Moment	Mc	0.288E+07	N-mm
Longitudinal Moment	Ml	0.375E+07	N-mm
Torsional Moment	Mt	0.432E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2225.5(24.0 - 3.0))}$$

$$= 881.682 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4451.000	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	6.601	mm
T / t	=	6.817	
d / t	=	25.492	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.376$$

Nr/P	=	0.097
Mr/P	=	0.069
M0/P	=	0.257
N0/P	=	0.379
MrD/Mc	=	0.084
NrDL/Mc	=	0.096
M0d/Mc	=	0.325
N0DL/Mc	=	-0.112
MrD/Ml	=	0.085
NrDL/Ml	=	0.097
M0D/Ml	=	0.310
N0DL/Ml	=	0.117

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	52	53	52	53

Outplane Stress Summary	39	66	83	28
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	21	29	35	20
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	39	67	83	28
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	19	37	50	3
-------------------------	----	----	----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53

Inplane Stress Summary	49	55	59	48
------------------------	----	----	----	----



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		49	55	60	48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		1	1	1	1
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-1	-1	1	1
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		14	15	14	15
Hoop Stress Summary		15	16	17	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		2	2	2	2
Axial Bending (P)		26	-26	26	-26
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-28	-28	28	28
Axial Bending (ML)		-112	112	112	-112
Axial Pressure Stress		7	7	7	7
Axial Stress Summary		-103	66	177	-100

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		4	4	-4	-4
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		14	14	14	14
Shear Stress Summary		18	18	10	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		124	72	178	120

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	15	14	15

Hoop Stress Summary	17	18	15	16
---------------------	----	----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-84	84	84	-84
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-69	45	143	-78
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	14	14	14	14

Shear Stress Summary	9	9	20	20
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	88	48	146	103
------------------------	----	----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{2213.5(24.0 - 3.0)}$$

$$= 879.756 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4427.000 mm
Nozzle Outside Diameter	(d) =	300.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	72.464 mm
T / t	=	0.290
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.984$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.107
Mr/P	=	0.229
M0/P	=	0.067
N0/P	=	0.123
MrD/Mc	=	0.311
NrDL/Mc	=	0.075



MØd/Mc = 0.096
 NØDL/Mc = 0.043
 MrD/Ml = 0.197
 NrDL/Ml = 0.093
 MØD/Ml = 0.062
 NØDL/Ml = 0.109

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	112	113	112	113
Outplane Stress Summary	110	114	137	100

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	52	59	124	-2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	110	115	137	102

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-12	12	12	-12
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	54	61	81	38

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-40	40	40	-40
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	113	112	113
Inplane Stress Summary	102	124	186	46

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	102	124	186	47

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	0	0	2	2	0	0	0	0
Circ. Q (SUS)	-13	13	27	-27	-3	3	5	-5
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-1	-1	1	1	1	1	1	1
Long. Q (SUS)	-4	4	6	-6	-9	9	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	52.0	53.0	54.0	55.0	52.0	53.0	52.0	53.0
Pm+Pl+Q (Total)	39.2	66.1	81.1	28.5	49.1	56.2	57.5	48.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	55.00	207.00	Passed
Pm+Pl+Q (TOTAL)	81.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	113	112	113	112	113	112	113
Circ. Pl (SUS)	-1	-1	5	5	1	1	3	3
Circ. Q (SUS)	-2	2	18	-18	-11	11	69	-69
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	4	4	2	2	2	2
Long. Q (SUS)	-4	4	62	-62	-4	4	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	112.0	113.0	112.0	113.0	112.0	113.0	112.0	113.0
Pm+Pl (SUS)	111.0	112.0	117.0	118.0	113.0	114.0	115.0	116.0
Pm+Pl+Q (Total)	109.0	114.0	135.1	102.0	102.0	125.0	184.0	47.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	118.00	207.00	Passed
Pm+Pl+Q (TOTAL)	184.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	15	14	15	14	15	14	15
Circ. Pl (SUS)	0	0	2	2	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-26	-26	30	30	-20	-20	24	24
Long. Q (SUS)	-86	86	138	-138	-58	58	110	-110
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-5	-5	5	5
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	14.0	15.0	14.0	15.0	14.0	15.0	14.0	15.0
Pm+Pl (SUS)	34.0	34.9	37.7	37.8	29.7	30.7	32.4	32.5
Pm+Pl+Q (Total)	124.3	72.6	175.6	119.7	87.9	47.6	143.8	102.3

Nozzle Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	118.00	Passed
Pm+Pl (SUS)	37.77	177.00	Passed
Pm+Pl+Q (TOTAL)	175.63	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N4-4''

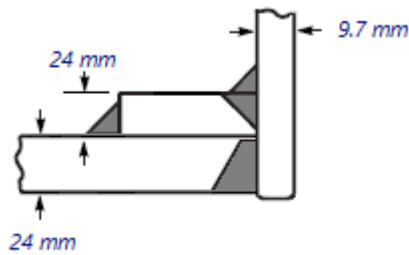
From: 20

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2950.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
 $= 17.2781 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.08 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 1.08)$
 $= 0.5198 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618 mm
Parallel to Vessel Wall, opening length	d	100.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		40.8364 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

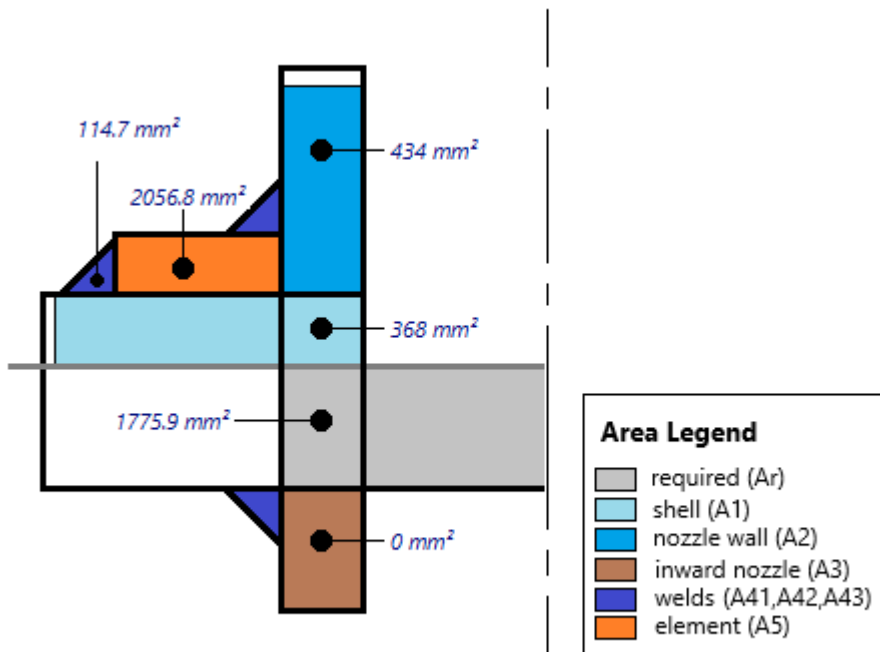


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1775.894	NA	NA
Area in Shell	A1	368.017	NA	NA
Area in Nozzle Wall	A2	434.011	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	114.729	NA	NA
Area in Element	A5	2056.800	NA	NA
TOTAL AREA AVAILABLE	Atot	2973.557	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	150.0974	24.0000 mm
Based on given Pad Diameter:	200.0000	10.0249 mm
Based on the Estimated Diameter Limit:	200.0743	10.0163 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (101 \cdot 17.3 \cdot 1 + 2 \cdot 6.73 \cdot 17.3 \cdot 1 \cdot (1 - 0.86))$
 $= 1775.894 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 101 (1 \cdot 21 - 1 \cdot 17.3) - 2 \cdot 6.73$
 $(1 \cdot 21 - 1 \cdot 17.3) \cdot (1 - 0.86)$
 $= 368.017 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 40.8) \cdot (6.73 - 0.52) \cdot 0.86$$

$$= 434.011 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100) \cdot 0.86 + (0) \cdot 0.86 + 29.2^2 \cdot 1$$

$$= 114.729 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr4}$$

$$= (200 - 114) \cdot 24 \cdot 1$$

$$= 2056.800 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5198 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 20.2781 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 20.2781 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(20.3, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.52, 8.26)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	257.7 MPa	Passed
Occasional	:	3.8,	Allowable	:	156.9 MPa	Passed
Shear	:	25.4,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 17.3, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.52, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.077$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.08/5.10 = 0.211$

Weld Size Calculations, Description: N4-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (1776 - 368 + 2 \cdot 6.73 \cdot 0.86 \cdot (1 \cdot 21 - 17.3)) \cdot 138)$$

$$= 200181.86 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (434 + 2057 + 115 - 0 \cdot 0.86) \cdot 138$$

$$= 359527.47 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (434 + 0 + 85.5 + (242)) \cdot 138$$

$$= 105059.10 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (434 + 0 + 115 + 2057 + (242)) \cdot 138$$

$$= 392900.41 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 103800. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 200 \cdot 18 \cdot 0.49 \cdot 138$$

$$= 382343. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 53.8) \cdot (9.73 - 3) \cdot 0.7 \cdot 118$$

$$= 93980. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 114 \cdot 22 \cdot 0.74 \cdot 138$$

$$= 403325. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 114 \cdot (24 - 3) \cdot 0.74 \cdot 138$$

$$= 384992. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (382343 + 93980) = 476323 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (103800 + 403325 + 384992 + 0) = 892117 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (382343 + 384992 + 0) = 767334 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 476322 N	, must exceed W = 200181 N	or W1 = 359527 N
Path 2-2 = 892116 N	, must exceed W = 200181 N	or W2 = 105059 N
Path 3-3 = 767334 N	, must exceed W = 200181 N	or W3 = 392900 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7424 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.7425 mm

Input Echo, WRC297 Item 1, Description: N4-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4400.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.08	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv , (4.762) is < 5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2225.5(24.0 - 3.0)}$
 $= 827.707 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4451.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	6.682	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.255$$

Nr/P	=	0.148
Mr/P	=	0.097
M0/P	=	0.306
N0/P	=	0.322
MrD/Mc	=	0.100
NrDL/Mc	=	0.122
M0d/Mc	=	0.314
N0DL/Mc	=	-0.144
MrD/Ml	=	0.101
NrDL/Ml	=	0.124
M0D/Ml	=	0.307
N0DL/Ml	=	-0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1



Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	52	53	52	53

Outplane Stress Summary	46	62	71	34
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	23	28	33	21
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	46	62	71	34

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	23	32	42	10
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53



Inplane Stress Summary	50	54	58	49
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	50	54	58	49
------------------------	----	----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	10	12	9	10
---------------------	----	----	---	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-75	75	75	-75
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-77	34	130	-58
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13

Shear Stress Summary	17	17	9	9
----------------------	----	----	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	95	44	131	71
------------------------	----	----	-----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	10	12	9	10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-58	58	58	-58
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-54	24	106	-47

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	13	13	13	13
Shear Stress Summary	8	8	19	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	67	28	110	69

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2213.5(24.0 - 3.0))}$$

$$= 825.781 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4427.000 mm
Nozzle Outside Diameter	(d) =	200.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	49.585 mm
T / t	=	0.424
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.656$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.127
Mr/P	=	0.286
M0/P	=	0.084
N0/P	=	0.153



MrD/Mc = 0.313
 NrDL/Mc = 0.075
 M0d/Mc = 0.098
 N0DL/Mc = -0.009
 MrD/Ml = 0.247
 NrDL/Ml = 0.087
 M0D/Ml = 0.079
 N0DL/Ml = 0.080

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	112	113	112	113
Outplane Stress Summary	111	115	132	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	53	59	112	6

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	111	115	132	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56

Outplane Stress Summary 57 59 74 42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-27	27	27	-27
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	113	112	113

Inplane Stress Summary 110 116 167 64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary 0 0 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 110 116 167 64

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	-8	8	18	-18	-2	2	4	-4
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	-3	3	5	-5	-5	5	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	53.0	54.0	53.0	54.0	52.0	53.0	52.0	53.0
Pm+Pl+Q (Total)	45.2	62.1	71.1	36.3	50.1	55.2	56.3	49.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed



Pm+Pl (SUS)	54.00	207.00	Passed
Pm+Pl+Q (TOTAL)	71.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	113	112	113	112	113	112	113
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	-1	1	15	-15	-3	3	51	-51
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	-3	3	51	-51	-1	1	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	112.0	113.0	112.0	113.0	112.0	113.0	112.0	113.0
Pm+Pl (SUS)	113.0	114.0	115.0	116.0	112.0	113.0	114.0	115.0
Pm+Pl+Q (Total)	112.0	115.0	130.0	101.0	109.0	116.0	165.0	64.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	116.00	207.00	Passed
Pm+Pl+Q (TOTAL)	165.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-26	-26	30	30	-20	-20	24	24
Long. Q (SUS)	-56	56	94	-94	-39	39	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-5	-5	5	5
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	32.6	33.5	34.4	34.4	27.9	28.8	29.3	29.4
Pm+Pl+Q (Total)	93.6	42.2	128.8	73.8	66.9	27.0	108.3	70.0

**Nozzle Stress Summation Comparison (MPa):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+Pl (SUS)	34.38	177.00	Passed
Pm+Pl+Q (TOTAL)	128.84	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K2B-2"

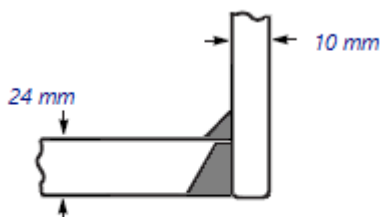
From: 20

Pressure for Reinforcement Calculations	P	1.1117	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.11 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.11)$
= 17.8342 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.11 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.11)$
= 0.2384 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.8000	mm
Parallel to Vessel Wall	Rn+tn+t	46.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2384	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 20.8342	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.8342	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(20.8, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.24, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 10, tr = 0.24, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.24, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1.11/1.96 = 0.567

Weld Size Calculations, Description: K2B-2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.1467 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3-2''

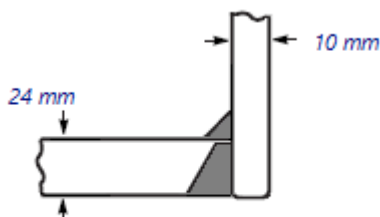
From: 30

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4050.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
= 17.2781 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.08 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.08)$
= 0.2310 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.8000	mm
Parallel to Vessel Wall	Rn+tn+t	46.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2310	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 20.2781	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.2781	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.42, max(20.3, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.23, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.08 / 1.96 = 0.550$

Weld Size Calculations, Description: K3-2''

Intermediate Calc. for nozzle/shell welds	Tmin	7.0000	mm
---	------	--------	----

**Results Per UW-16.1:**

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1467 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K4-2''

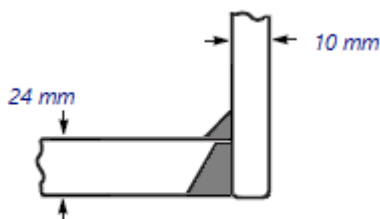
From: 30

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$$

$$= 17.2781 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 1.08 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.08)$$

$$= 0.2310 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 92.8000 mm

Parallel to Vessel Wall Rn+tn+t 46.4000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2310 mm

Wall Thickness per UG16(b), tr16b = 4.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 20.2781 mm

Wall Thickness tb1 = max(trb1, tr16b) = 20.2781 mm

Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm

Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(20.3, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.23, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.23, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 1.08/1.96 = 0.550$$

Weld Size Calculations, Description: K4-2''

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1467 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N5-4''

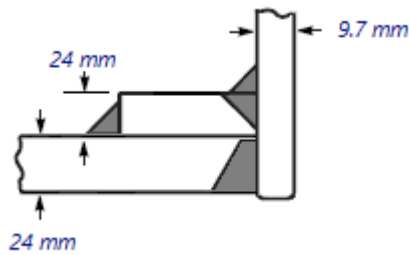
From: 40

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6650.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N5-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
 $= 17.2781 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.08 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 1.08)$
 $= 0.5198 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618 mm
Parallel to Vessel Wall, opening length	d	100.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		40.8364 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

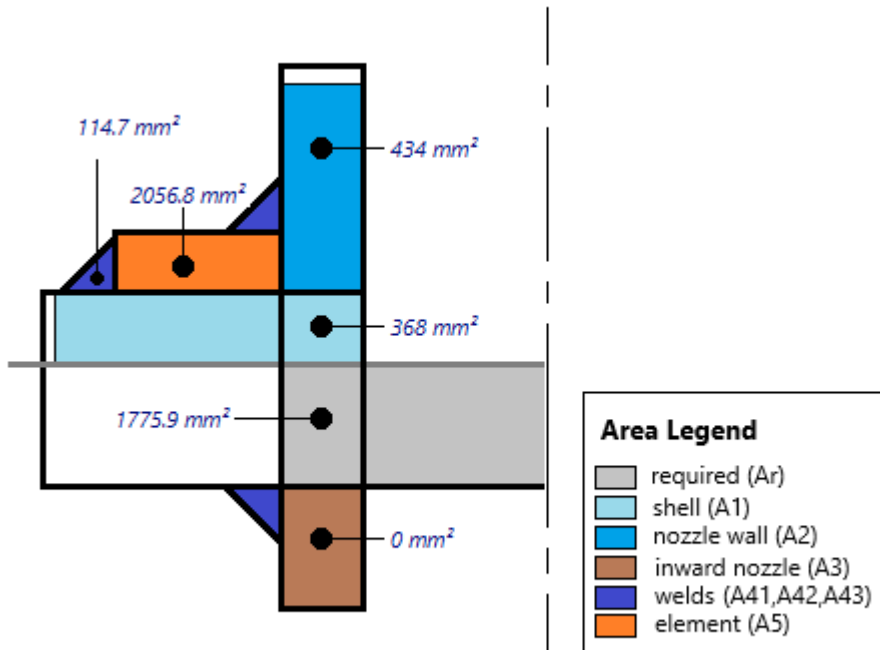


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1775.894	NA	NA
Area in Shell	A1	368.017	NA	NA
Area in Nozzle Wall	A2	434.011	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	114.729	NA	NA
Area in Element	A5	2056.800	NA	NA
TOTAL AREA AVAILABLE	Atot	2973.557	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	150.0974	24.0000 mm
Based on given Pad Diameter:	200.0000	10.0249 mm
Based on the Estimated Diameter Limit:	200.0743	10.0163 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (101 \cdot 17.3 \cdot 1 + 2 \cdot 6.73 \cdot 17.3 \cdot 1 \cdot (1 - 0.86))$
 $= 1775.894 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 101 (1 \cdot 21 - 1 \cdot 17.3) - 2 \cdot 6.73$
 $(1 \cdot 21 - 1 \cdot 17.3) \cdot (1 - 0.86)$
 $= 368.017 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 40.8) \cdot (6.73 - 0.52) \cdot 0.86$$

$$= 434.011 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100) \cdot 0.86 + (0) \cdot 0.86 + 29.2^2 \cdot 1$$

$$= 114.729 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot \text{fr4}$$

$$= (200 - 114) \cdot 24 \cdot 1$$

$$= 2056.800 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5198 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 20.2781 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 20.2781 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(20.3, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.52, 8.26)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	257.7 MPa	Passed
Occasional	:	3.8,	Allowable	:	156.9 MPa	Passed
Shear	:	25.4,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 17.3, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.52, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.077$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.08/5.10 = 0.211$

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (1776 - 368 + 2 \cdot 6.73 \cdot 0.86 \cdot (1 \cdot 21 - 17.3)) \cdot 138)$
 $= 200181.86 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (434 + 2057 + 115 - 0 \cdot 0.86) \cdot 138$
 $= 359527.47 \text{ N}$

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (434 + 0 + 85.5 + (242)) \cdot 138$
 $= 105059.10 \text{ N}$

Weld Load [W3]:

$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (434 + 0 + 115 + 2057 + (242)) \cdot 138$
 $= 392900.41 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118 \\
 &= 103800. \text{ N}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\
 &= (3.14/2.0) \cdot 200 \cdot 18 \cdot 0.49 \cdot 138 \\
 &= 382343. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 53.8) \cdot (9.73 - 3) \cdot 0.7 \cdot 118 \\
 &= 93980. \text{ N}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\
 &= (3.14/2) \cdot 114 \cdot 24 \cdot 0.74 \cdot 138 \\
 &= 439991. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 114 \cdot (24 - 3) \cdot 0.74 \cdot 138 \\
 &= 384992. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (382343 + 93980) = 476323 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\
 &= (103800 + 439991 + 384992 + 0) = 928783 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (S_{pew} + T_{ngw} + S_{inw}) \\
 &= (382343 + 384992 + 0) = 767334 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 476322 N	, must exceed W = 200181 N	or W1 = 359527 N
Path 2-2 = 928782 N	, must exceed W = 200181 N	or W2 = 105059 N
Path 3-3 = 767334 N	, must exceed W = 200181 N	or W3 = 392900 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7424 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.7425 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 100.83 mm

Input Echo, WRC297 Item 1, Description: N5-4'' :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4400.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	1.08	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (4.762) is < 5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2225.5(24.0 - 3.0))}$$

$$= 827.707 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4451.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	6.682	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.255$$

Nr/P	=	0.148
Mr/P	=	0.097
M0/P	=	0.306
N0/P	=	0.322
MrD/Mc	=	0.100
NrDL/Mc	=	0.122
M0d/Mc	=	0.314
N0DL/Mc	=	-0.144
MrD/Ml	=	0.101
NrDL/Ml	=	0.124
M0D/Ml	=	0.307
N0DL/Ml	=	-0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	52	53	52	53

Outplane Stress Summary	46	62	71	34
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top	Al Top	Bu Bottom	Bl Bottom
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	23	28	33	21
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top	Al Top	Bu Bottom	Bl Bottom
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au Top	Al Top	Bu Bottom	Bl Bottom
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	46	62	71	34

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left	Cl Left	Du Right	Dl Right
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	23	32	42	10
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu Left	Cl Left	Du Right	Dl Right
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53

Inplane Stress Summary	50	54	58	49
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	50	54	58	49
------------------------	----	----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	10	12	9	10
---------------------	----	----	---	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-75	75	75	-75
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-77	34	130	-58
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13

Shear Stress Summary	17	17	9	9
----------------------	----	----	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside



Two * Max Shear Stress 95 44 131 71

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses in the
hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary 10 12 9 10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-58	58	58	-58
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary -54 24 106 -47

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	13	13	13	13

Shear Stress Summary 8 8 19 19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 67 28 110 69

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2213.5(24.0 - 3.0)}$$

$$= 825.781 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4427.000 mm
Nozzle Outside Diameter	(d) =	200.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	49.585 mm
T / t	=	0.424
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.656$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P	=	0.127
Mr/P	=	0.286
M0/P	=	0.084
N0/P	=	0.153
MrD/Mc	=	0.313
NrDL/Mc	=	0.075
M0d/Mc	=	0.098
N0DL/Mc	=	-0.009
MrD/Ml	=	0.247
NrDL/Ml	=	0.087
M0D/Ml	=	0.079
N0DL/Ml	=	0.080

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	112	113	112	113
Outplane Stress Summary	111	115	132	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	53	59	112	6

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	111	115	132	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56

Outplane Stress Summary	57	59	74	42
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-27	27	27	-27
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	113	112	113

Inplane Stress Summary	110	116	167	64
------------------------	-----	-----	-----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	110	116	167	64
------------------------	-----	-----	-----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	-8	8	18	-18	-2	2	4	-4
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	-3	3	5	-5	-5	5	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	53.0	54.0	53.0	54.0	52.0	53.0	52.0	53.0
Pm+Pl+Q (Total)	45.2	62.1	71.1	36.3	50.1	55.2	56.3	49.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	54.00	207.00	Passed
Pm+Pl+Q (TOTAL)	71.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	113	112	113	112	113	112	113
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	-1	1	15	-15	-3	3	51	-51
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	-3	3	51	-51	-1	1	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	112.0	113.0	112.0	113.0	112.0	113.0	112.0	113.0
Pm+Pl (SUS)	113.0	114.0	115.0	116.0	112.0	113.0	114.0	115.0
Pm+Pl+Q (Total)	112.0	115.0	130.0	101.0	109.0	116.0	165.0	64.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	116.00	207.00	Passed
Pm+Pl+Q (TOTAL)	165.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-26	-26	30	30	-20	-20	24	24
Long. Q (SUS)	-56	56	94	-94	-39	39	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-5	-5	5	5
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	32.6	33.5	34.4	34.4	27.9	28.8	29.3	29.4



Pm+Pl+Q (Total)	93.6	42.2	128.8	73.8	66.9	27.0	108.3	70.0
-----------------	------	------	-------	------	------	------	-------	------

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+Pl (SUS)	34.38	177.00	Passed
Pm+Pl+Q (TOTAL)	128.84	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

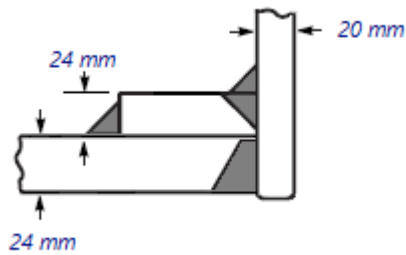
**Input, Nozzle Desc: N1-28"****From: 40**

Pressure for Reinforcement Calculations	P	1.1117	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6650.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		28.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1300.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		294.4000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-28"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	28.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.11 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.11)$
 $= 17.8342 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.11 \cdot 356 / (118 \cdot 1 + 0.4 \cdot 1.11)$
 $= 3.3378 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1354.4000 mm
Parallel to Vessel Wall, opening length	d	677.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		52.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

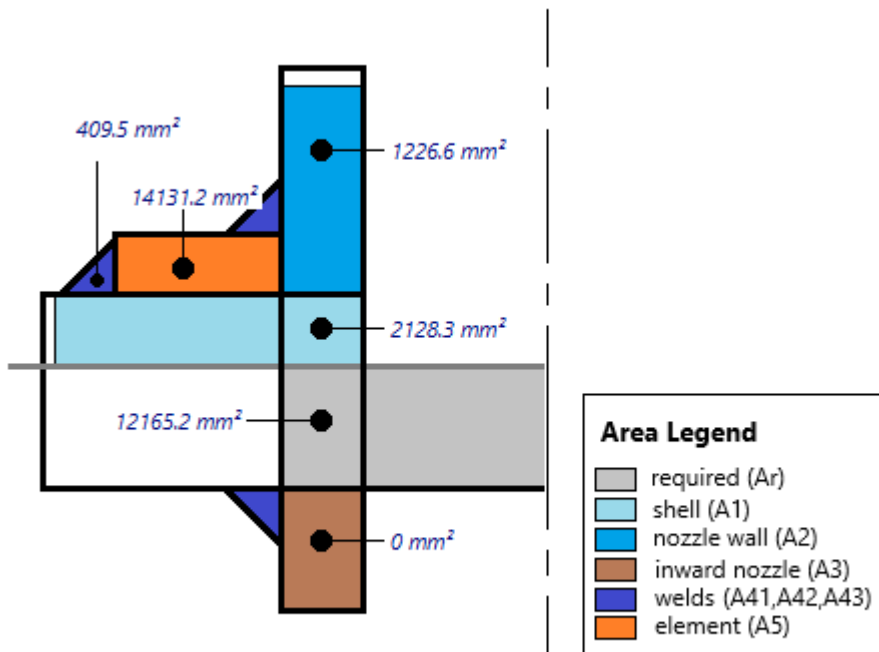


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12165.217	NA	NA
Area in Shell	A1	2128.264	NA	NA
Area in Nozzle Wall	A2	1226.628	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	409.507	NA	NA
Area in Element	A5	14131.199	NA	NA
TOTAL AREA AVAILABLE	Atot	17895.598	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1061.2340	24.0000 mm
Based on given Pad Diameter:	1300.0000	14.2677 mm
Based on Shell or Nozzle Thickness:	1131.2410	20.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (677 \cdot 17.8 \cdot 1 + 2 \cdot 17 \cdot 17.8 \cdot 1 \cdot (1 - 0.86))$
 $= 12165.217 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 677 (1 \cdot 21 - 1 \cdot 17.8) - 2 \cdot 17$
 $(1 \cdot 21 - 1 \cdot 17.8) \cdot (1 - 0.86)$
 $= 2128.264 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 52.5) \cdot (17 - 3.34) \cdot 0.86$$

$$= 1226.628 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 18^2 \cdot 1$$

$$= 409.507 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (1300 - 711) \cdot 24 \cdot 1$$

$$= 14131.199 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 6.3378 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 20.8342 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 20.8342 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(20.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(6.34, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.0,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	269.0 MPa	Passed
Occasional	:	10.8,	Allowable	:	156.9 MPa	Passed
Shear	:	10.8,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20$, $t_r = 3.34$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20$, $t_r = 3.34$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20$, $t_r = 3.34$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
--	-------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 17.8, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 20, tr = 3.34, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.11/1.96 = 0.567$

Weld Size Calculations, Description: N1-28"

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (12165 - 2128 + 2 \cdot 17 \cdot 0.86 \cdot (1 \cdot 21 - 17.8)) \cdot 138)$
 $= 1397656.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1227 + 14131 + 410 - 0 \cdot 0.86) \cdot 138$
 $= 2175667.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1227 + 0 + 85.5 + (611)) \cdot 138$
 $= 265299.28 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1227 + 0 + 410 + 14131 + (611)) \cdot 138$
 $= 2259911.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 711 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 645870. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 1300 \cdot 18 \cdot 0.49 \cdot 138$$

$$= 2485228. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 347) \cdot (20 - 3) \cdot 0.7 \cdot 118$$

$$= 1531048. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 711 \cdot 24 \cdot 0.74 \cdot 138$$

$$= 2737719. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 711 \cdot (24 - 3) \cdot 0.74 \cdot 138$$

$$= 2395504. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (2485227 + 1531047) = 4016274 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (645870 + 2737719 + 2395504 + 0) = 5779093 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (2485227 + 2395504 + 0) = 4880731 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4016274 N , must exceed W = 1397656 N or W1 = 2175667 N
 Path 2-2 = 5779093 N , must exceed W = 1397656 N or W2 = 265299 N
 Path 3-3 = 4880731 N , must exceed W = 1397656 N or W3 = 2259911 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 28.9292 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 452.9293 mm

Input Echo, WRC297 Item 1, Description: N1-28'' :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4400.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	711.200	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1300.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.11	MPa
Radial Load	P	-44798.03	N
Circumferential Shear	Vc	33598.50	N
Longitudinal Shear	Vl	44798.03	N
Circumferential Moment	Mc	0.627E+08	N-mm
Longitudinal Moment	Ml	0.816E+08	N-mm
Torsional Moment	Mt	0.941E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 711.2 + 2 \cdot 1.65 \cdot \sqrt{(2225.5(24.0 - 3.0))}$$

$$= 1424.607 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4451.000	mm
Nozzle Outside Diameter	(d) =	711.200	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	17.000	mm
T / t	=	2.647	
d / t	=	41.835	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.589$$

Nr/P	=	0.064
Mr/P	=	0.088
M0/P	=	0.064
N0/P	=	0.189
MrD/Mc	=	0.173
NrDL/Mc	=	0.100
M0d/Mc	=	0.153
N0DL/Mc	=	0.211
MrD/Ml	=	0.111
NrDL/Ml	=	0.071
M0D/Ml	=	0.089
N0DL/Ml	=	0.344

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	4	4	4	4
------------------------	---	---	---	---



Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	53	54	53	54

Outplane Stress Summary	16	61	116	39
-------------------------	----	----	-----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	-1	50	81	-17
------------------------	----	----	----	-----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	19	62	116	57

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	-9	53	88	-8
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-45	45	45	-45
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	54	53	54



Inplane Stress Summary	17	85	116	3
------------------------	----	----	-----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	27	85	116	13
------------------------	----	----	-----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	27	25	27

Hoop Stress Summary	10	11	49	50
---------------------	----	----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-180	180	180	-180
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-126	129	259	-205
----------------------	------	-----	-----	------

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	8	8	5	5
----------------------	---	---	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	138	129	259	256
------------------------	-----	-----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	27	25	27
Hoop Stress Summary	20	22	39	40

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-224	224	224	-224
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-168	176	301	-252

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	9	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	189	177	301	294

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 711.2 + 2 \cdot 1.65 \cdot \sqrt{(2213.5(24.0 - 3.0))}$$

$$= 1422.681 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4427.000 mm
Nozzle Outside Diameter	(d) =	1300.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	311.400 mm
T / t	=	0.067
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.264$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.079
M0/P	=	0.022
N0/P	=	0.028



MrD/Mc = 0.248
 NrDL/Mc = 0.106
 M0d/Mc = 0.074
 N0DL/Mc = 0.052
 MrD/Ml = 0.056
 NrDL/Ml = 0.059
 M0D/Ml = 0.017
 N0DL/Ml = 0.030

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	116	117	116	117
Outplane Stress Summary	113	116	151	96

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	55	54	167	-24

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	113	116	167	120

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary 19 90 128 4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-163	163	163	-163
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117

Inplane Stress Summary -5 225 344 -77

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary 0 0 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	25	225	344	81

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	54	53	54	53	54	53	54
Circ. Pl (SUS)	-15	-15	23	23	-3	-3	5	5
Circ. Q (SUS)	-22	22	38	-38	-34	34	56	-56
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-3	-3	5	5	-5	-5	13	13
Long. Q (SUS)	-26	26	48	-48	-31	31	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	53.0	54.0	53.0	54.0	53.0	54.0	53.0	54.0
Pm+Pl (SUS)	38.0	39.0	76.0	77.0	50.0	51.0	58.0	59.0
Pm+Pl+Q (Total)	19.4	61.3	114.1	56.1	26.3	85.1	114.1	11.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	54.00	138.00	Passed

Pm+Pl (SUS)	77.00	207.00	Passed
Pm+Pl+Q (TOTAL)	114.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	-2	-2	6	6	-6	-6	16	16
Circ. Q (SUS)	-1	1	27	-27	-115	115	211	-211
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-3	-3	13	13	-3	-3	7	7
Long. Q (SUS)	1	-1	95	-95	-35	35	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+Pl (SUS)	114.0	115.0	122.0	123.0	110.0	111.0	132.0	133.0
Pm+Pl+Q (Total)	113.0	116.0	166.1	120.0	25.0	226.0	343.0	82.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+Pl (SUS)	133.01	207.00	Passed
Pm+Pl+Q (TOTAL)	343.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	25	27	25	27	25	27	25	27
Circ. Pl (SUS)	-15	-15	23	23	-5	-5	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-11	-11	13	13	-8	-8	10	10
Long. Q (SUS)	-128	128	232	-232	-172	172	276	-276
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	25.0	27.0	25.0	27.0	25.0	27.0	25.0	27.0
Pm+Pl (SUS)	10.1	12.1	48.0	50.0	20.2	22.2	38.2	40.2
Pm+Pl+Q (Total)	137.7	129.4	257.1	257.2	188.2	176.1	298.2	294.4

**Nozzle Stress Summation Comparison (MPa):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	118.00	Passed
Pm+Pl (SUS)	50.04	177.00	Passed
Pm+Pl+Q (TOTAL)	298.25	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

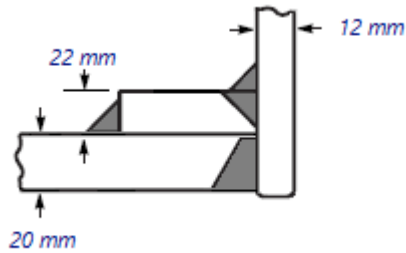
**Input, Nozzle Desc: N2-4''****Parent: N1-28''**

Pressure for Reinforcement Calculations	P	1.1117	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	671.20	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		49.2000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.11 \cdot 339 / (118 \cdot 1 - 0.6 \cdot 1.11)$
 $= 3.2083 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.11 \cdot 50.8 / (118 \cdot 1 + 0.4 \cdot 1.11)$
 $= 0.4768 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	167.2000 mm
Parallel to Vessel Wall, opening length	d	83.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: N2-4''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4768 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.2083 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.2083 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 8.0200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(6.21, 4.5)]$
 $= 6.2083 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.48, 6.21)$
 $= 6.2083 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**



Govrn. thk, tg = 12, tr = 0.48, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12, tr = 0.48, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12, tr = 0.48, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20, tr = 3.21, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12, tr = 0.48, c = 3 mm, E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.053$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.11/1.96 = 0.567$

Weld Size Calculations, Description: N2-4"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$8.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Expansion vessel (Electrolyzer Area)

PV Elite 28 Licensee: SL User



FileName : A162-V-XX01-Expansion vessel (Electrolyzer Area)- Page 159 of 185

Nozzle Calcs.: N2-4'' Nozl: 38 9:39am Sep 18,2026



HEXAGON

Converged Maximum Allowable Pressure in the Operating case: 5.75 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.8671 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8671 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N6-4''

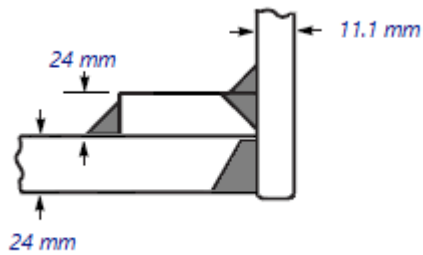
From: 60

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		11450.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N6-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
 $= 17.2781 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.08 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 1.08)$
 $= 0.5198 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		44.3130 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

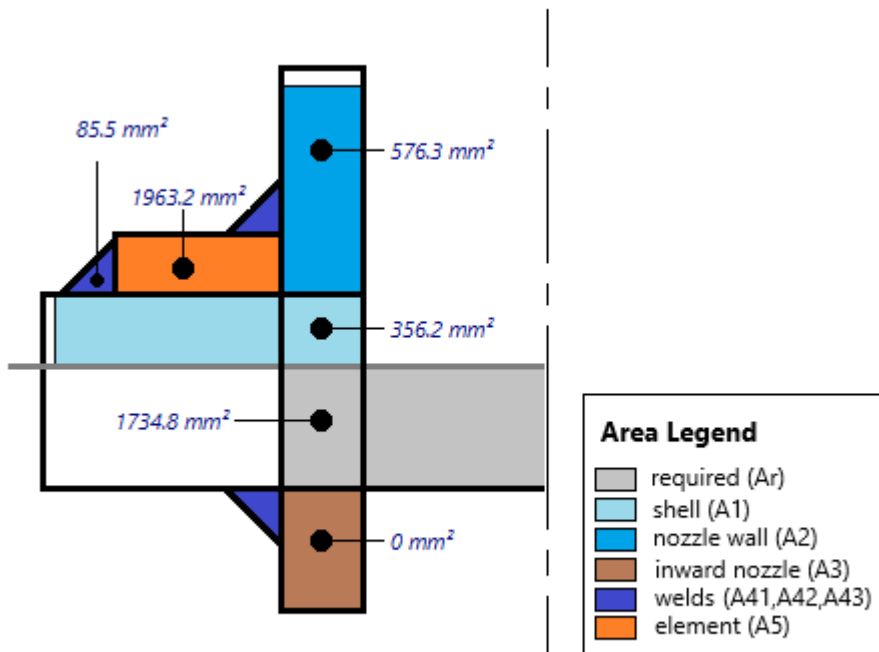


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1734.803	NA	NA
Area in Shell	A1	356.165	NA	NA
Area in Nozzle Wall	A2	576.346	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	1963.181	NA	NA
TOTAL AREA AVAILABLE	Atot	2981.199	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	144.1660	24.0000 mm
Based on given Pad Diameter:	200.0000	8.7627 mm
Based on Shell or Nozzle Thickness:	178.7289	11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (98 \cdot 17.3 \cdot 1 + 2 \cdot 8.13 \cdot 17.3 \cdot 1 \cdot (1 - 0.86))$
 $= 1734.803 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 98 (1 \cdot 21 - 1 \cdot 17.3) - 2 \cdot 8.13$
 $(1 \cdot 21 - 1 \cdot 17.3) \cdot (1 - 0.86)$
 $= 356.165 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 44.3) \cdot (8.13 - 0.52) \cdot 0.86$$

$$= 576.346 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$$

$$= 85.507 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$$

$$= (196 - 114) \cdot 24 \cdot 1$$

$$= 1963.181 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5198 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 20.2781 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 20.2781 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(20.3, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.52, 8.26)$$

$$= 8.2578 \text{ mm}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	31.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	263.1 MPa	Passed
Occasional	:	3.0,	Allowable	:	156.9 MPa	Passed
Shear	:	22.0,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.73, tr = 0.52, c = 3 \text{ mm, } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.077, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 17.3, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.52, c = 3 mm, E* = 1

Thickness Ratio = $tr * E*/(tg - c) = 0.077$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.08/5.10 = 0.211$

Weld Size Calculations, Description: N6-4"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (1735 - 356 + 2 \cdot 8.13 \cdot 0.86 \cdot (1 \cdot 21 - 17.3)) \cdot 138)$$

$$= 197368.66 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (576 + 1963 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 362217.38 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (576 + 0 + 85.5 + (292)) \cdot 138$$

$$= 131590.73 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (576 + 0 + 85.5 + 1963 + (292)) \cdot 138$$

$$= 402481.72 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118 \\
 &= 103800. \text{ N}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 200 \cdot 18 \cdot 0.49 \cdot 138 \\
 &= 382343. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 53.1) \cdot (11.1 - 3) \cdot 0.7 \cdot 118 \\
 &= 111921. \text{ N}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 114 \cdot 22 \cdot 0.74 \cdot 138 \\
 &= 403325. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 114 \cdot (24 - 3) \cdot 0.74 \cdot 138 \\
 &= 384992. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (382343 + 111921) = 494263 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (103800 + 403325 + 384992 + 0) = 892117 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (382343 + 384992 + 0) = 767334 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 494263 N	, must exceed W = 197368 N	or W1 = 362217 N
Path 2-2 = 892116 N	, must exceed W = 197368 N	or W2 = 131590 N
Path 3-3 = 767334 N	, must exceed W = 197368 N	or W3 = 402481 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7424 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.7425 mm

Input Echo, WRC297 Item 1, Description: N6-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4400.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	114.300	mm
Nozzle Thickness	11.1252	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.08	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.762) is < 5
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2225.5(24.0 - 3.0)}$
 $= 827.707 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4451.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	8.125	mm
T / t	=	5.538	
d / t	=	14.067	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.255$$

Nr/P	=	0.175
Mr/P	=	0.112
M0/P	=	0.295
N0/P	=	0.286
MrD/Mc	=	0.108
NrDL/Mc	=	0.142
M0d/Mc	=	0.306
N0DL/Mc	=	-0.137
MrD/Ml	=	0.108
NrDL/Ml	=	0.145
M0D/Ml	=	0.299
N0DL/Ml	=	-0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0



Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	52	53	52	53

Outplane Stress Summary	46	62	70	34
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	23	28	34	20
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	46	62	70	35

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	23	32	41	10
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53



Inplane Stress Summary	50	54	59	48
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	50	54	59	49
------------------------	----	----	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary	8	10	7	8
---------------------	---	----	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-24	-24	24	24
Axial Bending (ML)	-47	47	47	-47
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-51	14	92	-31
----------------------	-----	----	----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	14	14	8	8
----------------------	----	----	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	67	29	93	43
------------------------	----	----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8
Hoop Stress Summary	8	10	7	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-37	37	37	-37
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-36	10	77	-27

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11
Shear Stress Summary	7	7	15	15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	47	17	80	47

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2213.5(24.0 - 3.0))}$$

$$= 825.781 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4427.000 mm
Nozzle Outside Diameter	(d) =	200.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	50.975 mm
T / t	=	0.412
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.656$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.127
Mr/P	=	0.286
M0/P	=	0.084
N0/P	=	0.152



MrD/Mc = 0.313
 NrDL/Mc = 0.074
 M0d/Mc = 0.098
 N0DL/Mc = -0.007
 MrD/Ml = 0.247
 NrDL/Ml = 0.087
 M0D/Ml = 0.079
 N0DL/Ml = 0.080

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	112	113	112	113
Outplane Stress Summary	111	115	132	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	53	59	112	6

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	111	115	132	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56

Outplane Stress Summary 57 59 74 42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-27	27	27	-27
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	113	112	113

Inplane Stress Summary 110 116 167 64

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary 0 0 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 110 116 167 64

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-7	7	17	-17	-1	1	5	-5
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-2	2	6	-6	-5	5	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl+Q (Total)	45.2	60.1	69.1	36.3	51.1	54.2	57.2	48.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed



Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	69.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	113	112	113	112	113	112	113
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	-1	1	15	-15	-3	3	51	-51
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	-3	3	51	-51	-1	1	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	112.0	113.0	112.0	113.0	112.0	113.0	112.0	113.0
Pm+Pl (SUS)	113.0	114.0	115.0	116.0	112.0	113.0	114.0	115.0
Pm+Pl+Q (Total)	112.0	115.0	130.0	101.0	109.0	116.0	165.0	64.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	116.00	207.00	Passed
Pm+Pl+Q (TOTAL)	165.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	8	7	8	7	8	7	8
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-22	-22	26	26	-17	-17	21	21
Long. Q (SUS)	-33	33	61	-61	-23	23	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	7.0	8.0	7.0	8.0	7.0	8.0	7.0	8.0
Pm+Pl (SUS)	26.7	27.7	29.4	29.4	22.5	23.4	24.9	24.9
Pm+Pl+Q (Total)	65.3	28.6	90.8	43.1	46.2	15.5	78.2	46.1



Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	118.00	Passed
Pm+Pl (SUS)	29.42	177.00	Passed
Pm+Pl+Q (TOTAL)	90.76	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K1A-2"

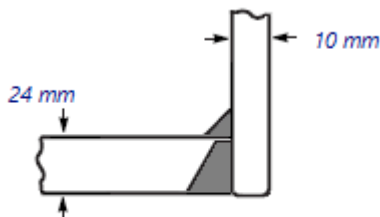
From: 60

Pressure for Reinforcement Calculations	P	1.0772	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12450.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.08 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.08)$
= 17.2781 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.08 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.08)$
= 0.2310 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	92.8000	mm
Parallel to Vessel Wall	Rn+tn+t	46.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2310	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 20.2781	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.2781	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(20.3, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.23, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.08 / 1.96 = 0.550$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
---	------	--------	----



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1467 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K1B-2"

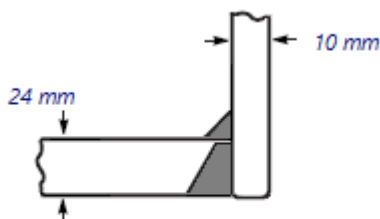
From: 60

Pressure for Reinforcement Calculations	P	1.1117	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	4400.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12450.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
---	-----------



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.11 \cdot 2203 / (138 \cdot 1 - 0.6 \cdot 1.11)$$

$$= 17.8342 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 1.11 \cdot 25.4 / (118 \cdot 1 + 0.4 \cdot 1.11)$$

$$= 0.2384 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 92.8000 mm

Parallel to Vessel Wall Rn+tn+t 46.4000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2384 mm

Wall Thickness per UG16(b), tr16b = 4.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 20.8342 mm

Wall Thickness tb1 = max(trb1, tr16b) = 20.8342 mm

Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm

Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.42, \max(20.8, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.24, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10, tr = 0.24, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.034, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 1.11/1.96 = 0.567$$

Weld Size Calculations, Description: K1B-2''

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1467 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.1468 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M2-24''

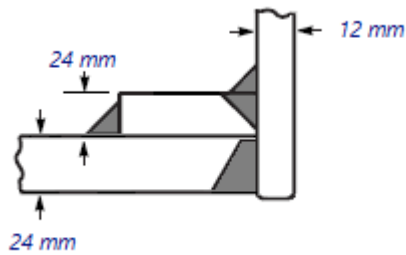
From: 70

Pressure for Reinforcement Calculations	P	1.0902	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	4400.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.09 \cdot 4406 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.09)$
 $= 15.6554 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.09 \cdot 305 / (118 \cdot 1 + 0.4 \cdot 1.09)$
 $= 2.8057 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001	mm
Parallel to Vessel Wall, opening length	d	591.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.5000	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118/138)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118/138)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, Sp/Sv)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.86, 1)$$

$$= 0.855$$

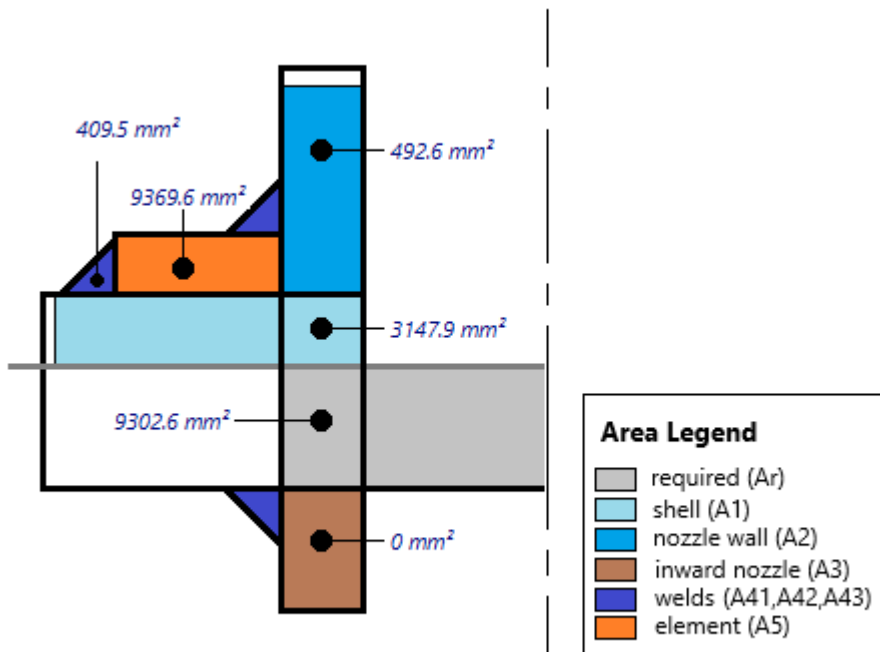


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9302.565	NA	NA
Area in Shell	A1	3147.933	NA	NA
Area in Nozzle Wall	A2	492.583	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	409.507	NA	NA
Area in Element	A5	9369.601	NA	NA
TOTAL AREA AVAILABLE	Atot	13419.623	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	828.4559	24.0000 mm
Based on given Pad Diameter:	1000.0000	13.4543 mm
Based on Shell or Nozzle Thickness:	1047.3119	12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (592 \cdot 15.7 \cdot 1 + 2 \cdot 9 \cdot 15.7 \cdot 1 \cdot (1 - 0.86))$
 $= 9302.565 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 592 (1 \cdot 21 - 1 \cdot 15.7) - 2 \cdot 9$
 $(1 \cdot 21 - 1 \cdot 15.7) \cdot (1 - 0.86)$
 $= 3147.933 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 46.5) \cdot (9 - 2.81) \cdot 0.86$$

$$= 492.583 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 18^2 \cdot 1$$

$$= 409.507 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (1000 - 610) \cdot 24 \cdot 1$$

$$= 9369.601 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.8057 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.8057 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 12$, $t_r = 2.81$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.81$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.81$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 24$, $t_r = 15.7$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12$, $t_r = 2.81$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.09/1.96 = 0.556

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.3000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5 * TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9303 - 3148 + 2 \cdot 9 \cdot 0.86 \cdot (1 \cdot 21 - 15.7)) \cdot 138)$
 $= 860602.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (493 + 9370 + 410 - 0 \cdot 0.86) \cdot 138$
 $= 1417347.12 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (493 + 0 + 85.5 + (323)) \cdot 138$
 $= 124367.64 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (493 + 0 + 410 + 9370 + (323)) \cdot 138$
 $= 1461946.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 118$
 $= 553603. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 1000 \cdot 18 \cdot 0.49 \cdot 138$
 $= 1911713. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 300) \cdot (12 - 3) \cdot 0.7 \cdot 118$
 $= 701266. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 610 \cdot 22 \cdot 0.74 \cdot 138$
 $= 2151065. \text{ N}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 610 \cdot (24 - 3) \cdot 0.74 \cdot 138$$

$$= 2053289. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1911713 + 701266) = 2612979 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (553603 + 2151065 + 2053289 + 0) = 4757956 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1911713 + 2053289 + 0) = 3965002 \text{ N}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 2612979 \text{ N}, \text{ must exceed } W = 860602 \text{ N} \quad \text{or } W1 = 1417347 \text{ N}$$

$$\text{Path 2-2} = 4757956 \text{ N}, \text{ must exceed } W = 860602 \text{ N} \quad \text{or } W2 = 124367 \text{ N}$$

$$\text{Path 3-3} = 3965002 \text{ N}, \text{ must exceed } W = 860602 \text{ N} \quad \text{or } W3 = 1461946 \text{ N}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.08 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.6908 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.6909 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.